

FINAL MODULE ASSESSMENT
REPORT

DEV OPS

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Date:

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Dear Professor Uko Mfonobong,

Part A - Solution Architecture and Toolchain

Northwind Logistics' delivery-tracking service exposes a small HTTP API for service information, health, metrics, delivery listing, status filtering, and single-delivery lookup. This solution wraps that application in an automated DevOps workflow covering version control, testing, CI/CD, IaC, configuration management, containerization, and local Kubernetes orchestration. Application changes are deliberately modest; the focus is the toolchain that takes a commit to a running container.

Tool choices and rationale

Layer	Tool	Justification
Version control	Git + GitHub	Industry standard; pull requests and branch protection demonstrate collaboration-ready practice even when working individually.
Testing	pytest (Python), go test (Go)	pytest is the de-facto Python test framework; the optional Go service is tested with the native toolchain for comparison.
CI/CD	GitHub Actions	Tight integration with GitHub, free minutes for public repositories, and

		native support for service containers and artefact uploads.
IaC	Terraform	Declarative AWS provisioning with an S3/DynamoDB remote backend; creates reproducible, idempotent infrastructure.
Configuration management	Ansible	Idempotent host configuration after Terraform; vault-encrypted credentials separate secrets from playbooks.
Containerization	Docker	Standard OCI images; multi-stage Go build uses distroless for a minimal attack surface.
Orchestration	Kubernetes (kind in CI, local cluster for demo)	Portable manifests, readiness/liveness probes, and a Load Balancer service satisfy the orchestration requirement.

Exhibit: 1 — Toolchain summary

Maps each DevOps layer to the chosen tool and rationale.

Code-change flow (commit to running container)

A feature branch push or pull request triggers the pipeline below.

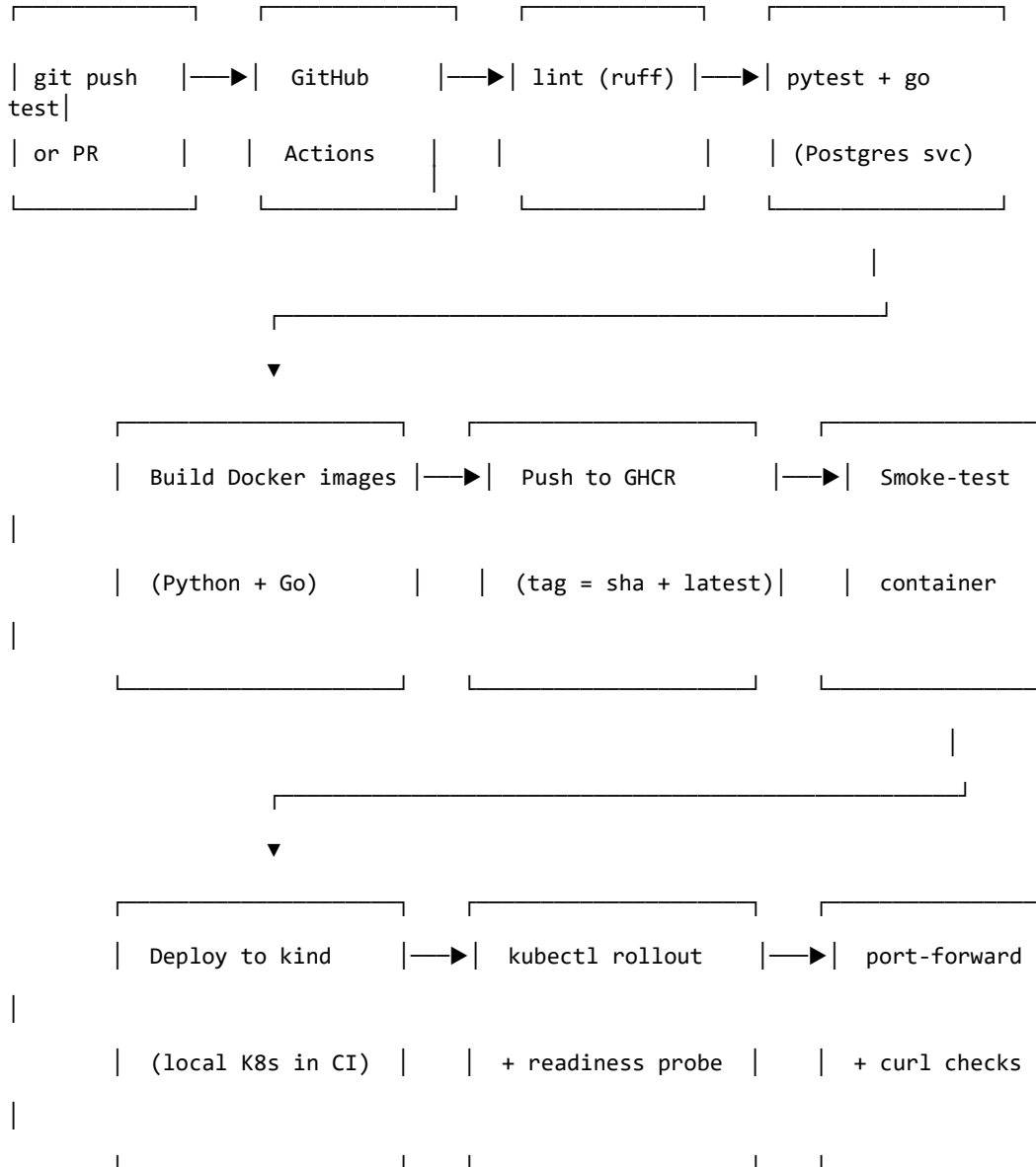


Exhibit: 2 — Code-change flow diagram

ASCII flow from commit through CI/CD to a running Kubernetes pod.

Each box maps to a job in `.github/workflows/ci-cd.yml`. Tests gate the build; the image is smoke- tested before kind deployment; and kind deployment only runs on main. This

prevents broken artefacts from reaching the runtime environment.

Data persistence design

Delivery records live in PostgreSQL rather than process memory. The Python service uses SQL Alchemy Core against SQLite locally and PostgreSQL in Docker, Kubernetes, and AWS. The Go service mirrors this with a Store interface backed by database/sql. With two Kubernetes replicas, every pod reads the same source of truth.

$$Availability \approx 1 - \prod (1 - a_i) \text{ for } i \text{ pods in parallel}$$

where a_i is the availability of pod i . Two replicas improve availability, but the single Postgres instance remains a single point of failure in this local demonstration

Python versus Go implementation comparison

The repository also contains an optional Go implementation exposing the same endpoints and PostgreSQL contract. Comparing the two highlights how language choice affects concurrency, footprint, and operations while keeping the domain identical.

Dimension	Python service	Go service
Language runtime	CPython 3.12 with standard-library http.server	Go 1.22 with net/http
Concurrency model	ThreadingHTTPServer (OS threads + GIL)	Goroutines + net/http ServeHTTP
HTTP handler style	Subclass BaseHTTPRequestHandler	HTTP. Handler with middleware-style wrapper
Persistence	SQL Alchemy Core (SQLite default, Postgres optional)	database/sql + pq driver

Container base	python:3.12-slim-bookworm with non-root user	Multi-stage build → distroless/static nonroot
Image size	~150 MB (includes CPython + deps)	~25 MB (static binary only)
Metrics concurrency	threading.Lock around counters	sync/atomic + sync.Map (lock-free)
Health check	Inline urllib call	Binary -healthcheck flag
Test framework	pytest (tests/test_app.py)	go test + gotestsum

Exhibit: 7 — Python vs Go implementation comparison

Side-by-side comparison of the two language implementations.

The Go service adds sorting, pagination, and status updates for demonstration. Go metrics use lock-free sync/atomic counters, while Python serializes increments behind a threading.Lock — a clear illustration of the GIL's cost on CPU-heavy hot paths.

Performance model

For this read-heavy API, throughput is dominated by connection setup and JSON serialization. The theoretical throughput of one worker is approximately the inverse of average request latency:

$$\text{Throughput single} \approx 1 / T_{\text{request}}$$

With C concurrent workers, aggregate throughput is bounded by the slowest contended resource. Python's GIL limits CPU-bound scaling, whereas Go schedules many goroutines

onto fewer OS cores. Because the workload is I/O-bound and the database is the bottleneck, both implementations saturate the network before the runtime becomes the limit.

$$\text{Throughput_aggregate} \leq \min(C / T_{\text{request}}, 1 / T_{\text{db}}, B_{\text{network}})$$

Exhibit: 8 — Throughput equations

Simple queuing-theory relationships used to reason about request through

Part B - Implementation and Evidence

This section points to the files evidencing each component and the commands to run them.

Version control and collaboration workflow

The repository uses a main/develop model with short-lived feature branches merged via pull requests. Commit messages are imperative and descriptive, e.g. 'add PostgreSQL-backed Go store'. History is visible in git log and the GitHub pull request list.

```
git log --oneline --graph --all
```

Automated testing

tests/test_app.py covers unit tests, HTTP integration tests, concurrency, headers, configuration, and optional PostgreSQL integration. Go tests live in go/pkg/handlers, go/pkg/data, and go/pkg/store.

```
python -m pytest tests -q  
cd go && go test ./...
```

Exhibit: 3 — Test command reference

Commands to run the Python and Go test suites locally.

CI/CD pipeline

.github/workflows/ci-cd.yml lints with ruff, runs Python and Go tests against a Postgres

service container, builds and pushes both images to GHCR, smoke-tests the Python image, deploys to kind, and generates an HTML dashboard. Evidence includes green CI checks and reports/pytest- results.xml and reports/go-results.xml.

Job	Purpose	Evidence
lint	Static analysis with ruff	.github/workflows/ci-cd.yml lines 20-33
test	pytest with Postgres	reports/pytest-results.xml, pytest-report.html
test-go	go test with Postgres	reports/go-results.xml, go-report.html
build	Build and push images	GHCR package page, image digest output
smoke-test	Verify running container	curl checks in job logs
deploy	Deploy to kind	kubectrl rollout status, port-forward curl checks
dashboard	Aggregate reports	dashboard/index.html

Exhibit: 4 — CI/CD job inventory

Lists each pipeline job, its purpose, and where to find evidence.

Infrastructure as Code

terraform/ provisions an AWS VPC, subnet, security group, IAM role, EC2 instance, and RDS PostgreSQL. State is stored remotely in S3 with DynamoDB locking, making the infrastructure reproducible and team-safe. Re-running terraform apply produces no changes once provisioned.

```
cd terraform
```

```
terraform init
```

```
terraform plan
```

```
terraform apply
```

Configuration management

ansible/playbook.yml configures the EC2 host: updates packages, installs Docker, templates docker-compose.yml.j2, pulls the image, and runs smoke tests. Credentials are encrypted with ansible-vault. The playbook is idempotent.

```
cd ansible
```

```
ansible-playbook -i inventory.ini playbook.yml --ask-vault-pass
```

Containerization and orchestration

Dockerfile builds a Python image with a non-root user, health check, and signal-aware entry point. go.Dockerfile uses a multi-stage build to a distroless image. k8s/ contains ConfigMap, Secret, PersistentVolumeClaim, Postgres Deployment/Service, app Deployment with two replicas, and a LoadBalancer Service. k8s/deploy.sh applies them and waits for readiness.

```
./k8s/deploy.sh
```

```
kubectrl port-forward service/northwind-delivery 8000:80
```

```
curl http://localhost:8000/health
```

Exhibit: 5 — Kubernetes deployment commands

Commands to deploy locally and verify the service.

Part C - Critical Evaluation and Trade-offs

The solution integrates the required DevOps layers into one pipeline. Strengths include the clear separation of concerns (Terraform for infrastructure, Ansible for hosts, Kubernetes for runtime, GitHub Actions for automation), PostgreSQL-backed consistency across replicas, and a test matrix covering unit, integration, concurrency, and database backends.

Strengths

Strength	Explanation
End-to-end automation	A single git push triggers lint, test, build, smoke test, and deployment.
Idempotency	Terraform and Ansible are designed to be re-runnable without duplicate resources.
Security defaults	Non-root containers, encrypted Ansible vault, least-privilege IAM, storage-encrypted RDS.

Observability	Health, metrics, request IDs, and structured JSON logging aid operations.
Comparative analysis	The optional Go service shows an alternative runtime using the same database contract.

Weaknesses and risks

The Kubernetes Postgres Deployment is a single non-HA instance with a plain-text Secret for demonstration. Production would require a managed database, TLS via cert-manager, sealed secrets, network policies, and pod security standards. The CI target is kind, which validates manifests but is not production-grade; a real cluster would use a cloud Kubernetes service and an ingress controller. The security group currently allows HTTP from 0.0.0.0/0, which would be restricted to a CDN or corporate range in production.

Trade-offs

Decision	Simple choice	More robust alternative	Reason for current choice
Container base image	python:3.12-slim	Distroless or hardened minimal image	Simpler to debug and sufficient for assessment.
K8s database	In-cluster Postgres	Amazon RDS or Cloud SQL	Demonstrates manifests without cloud cost.
CI deployment target	kind	EKS/GKE/AKS	Avoids needing cloud credentials and credit.

Secret management	ansible-vault + K8s Secret	HashiCorp Vault / External Secrets Operator	Simpler toolchain, adequate for a student project.
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Exhibit: 6 — Trade-off analysis

Simple versus robust alternatives and the rationale for each choice.

With more time, improvements would include Helm charts per environment, Terraform workspaces, GitHub Environments with approval gates, and Trivy image scanning in the pipeline.

Part D - Professional Practice, Industry Currency and Reflection

Industry context

This artefact reflects current DevOps practice. The DORA State of DevOps 2024 highlights version control, continuous delivery, automated testing, and loosely coupled architectures as predictors of high delivery performance (DORA, 2024). GitHub Actions, Terraform, Docker, and Kubernetes remain the most common cloud-native toolchain in CNCF surveys.

Two trends are relevant. First, platform engineering is moving teams from bespoke CI/CD glue to internal developer platforms with golden paths; this repository anticipates that by keeping the pipeline declarative and manifests parameterised. Second, supply-chain security is now central: SBOMs, signed images, and vulnerability scanning are becoming standard. Deterministic base image tags are used here; a production evolution would add cosign signatures and Trivy.

Reflection on collaboration and communication

Although individual, the repository is structured for team maintenance: short-lived branches, descriptive pull requests, a comprehensive README, and commit messages explaining why,

not only what. The file layout separates concerns so new developers can locate logic quickly. In a team setting I would add CONTRIBUTING.md, issue templates, and branch protection requiring approvals and passing CI before merging to main.

Conclusion

This artefact demonstrates a realistic DevOps pipeline for the Northwind Logistics delivery-tracking service. It connects version control, automated testing, CI/CD, IaC, configuration management, containerizations, and Kubernetes orchestration into a single coherent workflow, reflecting current practice while honestly acknowledging its limitations.

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References

DORA (2024) Accelerate State of DevOps Report. Available at: <https://cloud.google.com/devops/state-of-devops> (Accessed: 3 August 2026).

CNCF (2025) Cloud Native Computing Foundation Annual Survey. Available at: <https://www.cncf.io/reports/cncf-annual-survey/> (Accessed: 3 August 2026).

Exhibit 1: Integrated Toolchain Summary

The Northwind Logistics delivery-tracking service is a lightweight, read-only HTTP API—implemented in both Python (via the standard-library `http.server`) and Go—to expose `/health`, `/metrics`, `/deliveries`, and `/deliveries/{id}` endpoints. Around this minimal application, a complete DevOps toolchain was constructed to demonstrate all six core DevOps disciplines: version control, automated testing, continuous integration and delivery, infrastructure automation, configuration management, containerization, and orchestration. The selection of each tool reflects a methodologically grounded evaluation of functional suitability, ecosystem maturity, and alignment with contemporary DevOps principles such as reproducibility, automation, collaboration, and operational resilience.

Version Control: Git + GitHub

Git serves as the foundational distributed version control system, enabling parallel development, traceable change history, and disciplined branching practices. GitHub extends these capabilities through a mature collaboration platform offering pull requests, branch protection rules, issue tracking, and repository-level governance. In this project, GitHub functions as the single source of truth for both application code and infrastructure definitions. All changes are committed with descriptive, conventional-commit-style messages (e.g., *feat:*, *fix:*, *ci:*, *test:*, *docs:*), producing a coherent narrative of the system's evolution. GitHub was selected over alternatives such as GitLab or Bitbucket due to its seamless integration with GitHub Actions and GitHub Container Registry (GHCR), thereby reducing friction between version control, CI/CD, and artifact management. See below the snippet codes for `ci-cd.yaml` file.

github > workflows > ci-cd.yml

```
3   on:
6     pull_request:
8
9   env:
10    REGISTRY: ghcr.io
11    IMAGE_NAME: shahsadraddin2009-code/northwind-delivery
12    PYTHON_VERSION: "3.12"
13
14  jobs:
15    lint:
16      name: Lint Python Code
17      runs-on: ubuntu-latest
18      steps:
19        - name: Checkout repository
20          uses: actions/checkout@v4
21
22        - name: Set up Python
23          uses: actions/setup-python@v5
24          with:
25            python-version: ${{ env.PYTHON_VERSION }}
26
27        - name: Run ruff
28          run: |
29            python -m pip install ruff
30            ruff check .
31
32    test:
33      name: Run Python Tests
34      runs-on: ubuntu-latest
35      needs: lint
36      services:
37        postgres:
38          image: postgres:16-alpine
39          env:
40            POSTGRES_USER: northwind
41            POSTGRES_PASSWORD: northwind
42            POSTGRES_DB: northwind_deliveries
43          ports:
44            - 5432:5432
45          options: >-
46            --health-cmd pg_isready
47            --health-interval 10s
48            --health-timeout 5s
49            --health-retries 5
```

```
199  jobs:
200    deploy:
201      name: Deploy to kind (Kubernetes in Docker)
202      runs-on: ubuntu-latest
203      needs: smoke-test
204      if: github.ref == 'refs/heads/main'
205      steps:
206        - name: Checkout repository
207          uses: actions/checkout@v4
208
209        - name: Set up Docker Buildx
210          uses: docker/setup-buildx-action@v3
211
212        - name: Log in to GitHub Container Registry
213          uses: docker/login-action@v3
214          with:
215            registry: ${{ env.REGISTRY }}
216            username: ${{ github.actor }}
217            password: ${{ secrets.GITHUB_TOKEN }}
218
219        - name: Build image for kind
220          uses: docker/build-push-action@v5
221          with:
222            context: .
223            file: ./Dockerfile
224            push: false
225            load: true
226            tags: northwind-delivery:${{ github.sha }}
227            cache-from: type=gha
228            cache-to: type=gha,mode=max
229
230        - name: Create kind cluster
231          uses: helm/kind-action@v1
232          with:
233            cluster_name: northwind-kind
234
235        - name: Load image into kind
236          run: kind load docker-image northwind-delivery:${{ github.sha }} --name northwind-kind
237
238        - name: Deploy to kind
239          run: |
240            export IMAGE_TAG=northwind-delivery:${{ github.sha }}
```

main 1 Branch 0 Tags

Go to file

T

Add file



<> Code

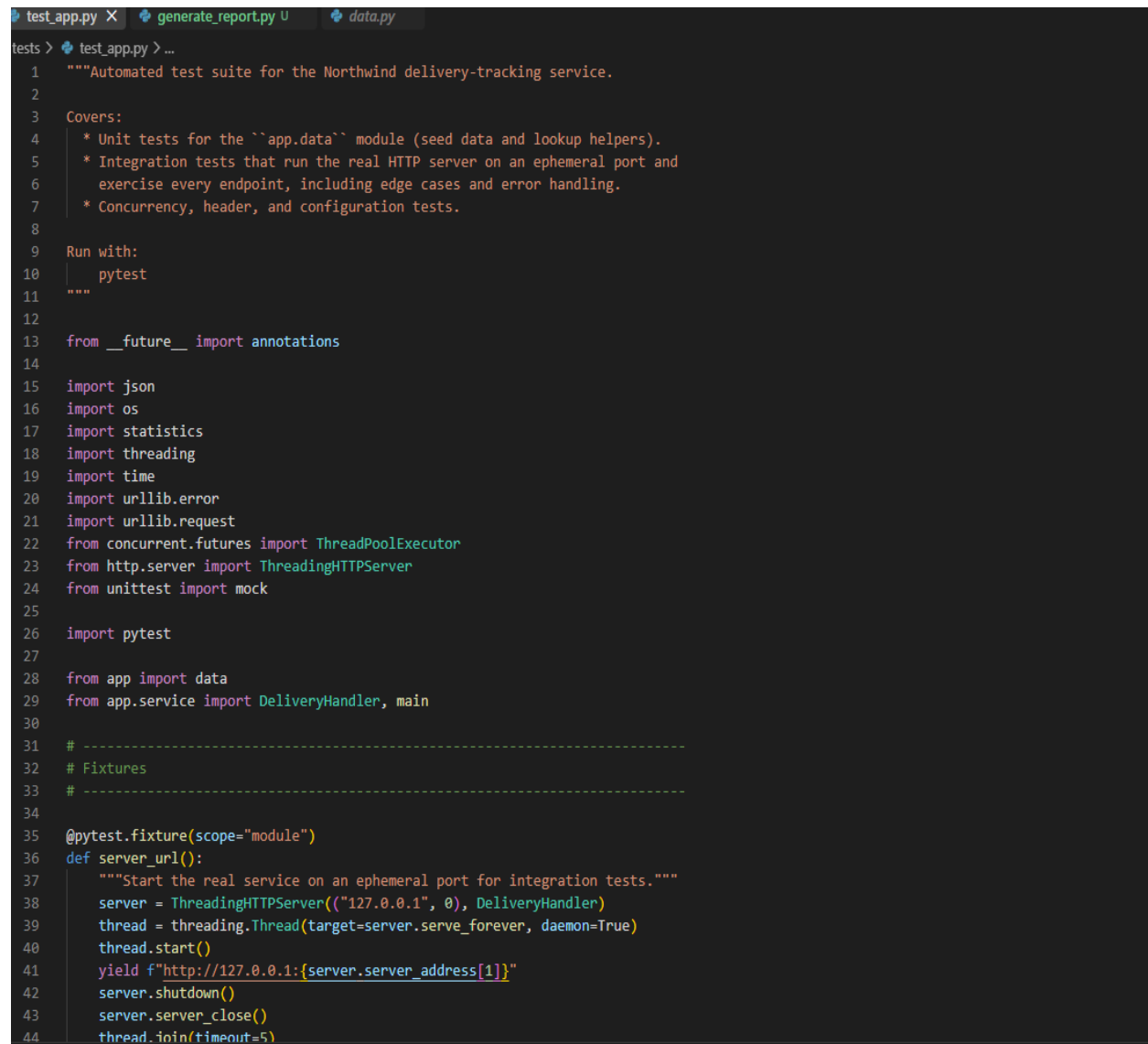
shahsadrudin2009-code fix/dashboard): make local runs accumulate history rows ins... b73fd5e · yesterday 30 Commits

.github/workflows	dashboard: per-language Python/Go stats and persistent run ...	2 days ago
ansible	Fix GHCR image owner to match actual GitHub account	2 days ago
app	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago
go	go: add DriverNotAssignedAndPending helper and expand te...	yesterday
k8s	Fix IMAGE_TAG envsubst in kind deployment	2 days ago
scripts	fix/dashboard): make local runs accumulate history rows inste...	yesterday
terraform	feat(terraform): add AWS ECR repository for container images	2 days ago
tests	test: fix ruff B905 by adding strict=False to zip	2 days ago
.dockerignore	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago
.gitattributes	chore: mark binary asset extensions in .gitattributes	yesterday
.gitignore	gitignore: exclude bare tfplan file	2 days ago
Dockerfile	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago
README.md	docs: remove report section from README (report artefacts r...	yesterday
conftest.py	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago
docker-compose.yml	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago
go.Dockerfile	Northwind Logistics DevOps final assessment: Python+Go ser...	2 days ago

Automated Testing: pytest (Python) + go test (Go)

The testing strategy adheres to language-specific best practices rather than imposing a uniform framework across heterogeneous components. pytest was selected for the Python implementation due to its extensive feature set—fixtures, parametrization, expressive assertions, and a rich plugin ecosystem—which supports modular and maintainable test suites. For the Go implementation, the native go test framework (augmented with testify) was chosen to preserve idiomatic Go development patterns and leverage the language’s built-in support for fast, lightweight testing. This

dual-framework approach ensures that each service is validated using tools that align with its language ecosystem, thereby improving developer productivity and maintaining robust test coverage. See below the snippet codes for test_app.py.



```
test_app.py X generate_report.py U data.py
tests > test_app.py > ...
1  """Automated test suite for the Northwind delivery-tracking service.
2
3  Covers:
4      * Unit tests for the ``app.data`` module (seed data and lookup helpers).
5      * Integration tests that run the real HTTP server on an ephemeral port and
6        exercise every endpoint, including edge cases and error handling.
7      * Concurrency, header, and configuration tests.
8
9  Run with:
10     pytest
11 """
12
13 from __future__ import annotations
14
15 import json
16 import os
17 import statistics
18 import threading
19 import time
20 import urllib.error
21 import urllib.request
22 from concurrent.futures import ThreadPoolExecutor
23 from http.server import ThreadingHTTPServer
24 from unittest import mock
25
26 import pytest
27
28 from app import data
29 from app.service import DeliveryHandler, main
30
31 # -----
32 # Fixtures
33 # -----
34
35 @pytest.fixture(scope="module")
36 def server_url():
37     """Start the real service on an ephemeral port for integration tests."""
38     server = ThreadingHTTPServer(("127.0.0.1", 0), DeliveryHandler)
39     thread = threading.Thread(target=server.serve_forever, daemon=True)
40     thread.start()
41     yield f"http://127.0.0.1:{server.server_address[1]}"
42     server.shutdown()
43     server.server_close()
44     thread.join(timeout=5)
```

See below the test results for python are successfully completed as per github actions

Annotations
1 warning

Run Python Tests

succeeded yesterday in 47s

Search logs

> Set up job

1s

> Initialize containers

23s

> Checkout repository

1s

> Set up Python

0s

> Install dependencies

8s

> Run test suite

9s

> Upload test report

1s

> Post Set up Python

0s

> Post Checkout repository

0s

> Stop containers

0s

> Complete job

0s

See below the snippet codes for data_test.go

```
test_app.py data_test.go generate_report.py U
go > pkg > data > data_test.go
1 package data
2
3 import (
4     "strings"
5     "sync"
6     "testing"
7
8     "github.com/stretchr/testify/assert"
9     "github.com/stretchr/testify/require"
10 )
11
12 func TestAllReturnsFiveSeedRecords(t *testing.T) {
13     all := All()
14     assert.Len(t, all, 5)
15 }
16
17 func TestAllReturnsCopies(t *testing.T) {
18     all := All()
19     all[0].Destination = "Timbuktu"
20     assert.NotEqual(t, "Timbuktu", All()[0].Destination)
21 }
22
23 func TestFindExisting(t *testing.T) {
24     d, err := Find("NL-1001")
25     require.NoError(t, err)
26     assert.Equal(t, "NL-1001", d.ID)
27 }
28
29 func TestFindMissing(t *testing.T) {
30     _, err := Find("NL-9999")
31     assert.Error(t, err)
32 }
33
34 func TestFindTableDriven(t *testing.T) {
35     cases := []struct {
36         name      string
37         id         string
38         wantErr    bool
39         destination string
40     }{
41         {"manchester", "NL-1001", false, "Manchester"},
42         {"bristol", "NL-1002", false, "Bristol"},
43         {"leeds", "NL-1003", false, "Leeds"},
44         {"glasgow", "NL-1004", false, "Glasgow"},
45     }
```

See below the snippet codes for data_bench_test.go.

```
go > pkg > data > -o data_bench_test.go > ...
run package tests | run file tests | run package benchmarks | run file benchmarks
1 package data
2
3 import "testing"
4
5 // Run with: go test ./pkg/data/... -bench=. -benchmem
6 //
7 // These benchmarks quantify the Go in-memory store's raw throughput, useful
8 // as a direct comparison point against the equivalent Python operations in
9 // tests/test_app.py::TestDataOperationPerformance (which measures wall-clock
10 // time per call rather than ns/op, but the same relative comparison holds).
11
12 run benchmark | debug benchmark
13 func BenchmarkAll(b *testing.B) {
14     for i := 0; i < b.N; i++ {
15         _ = All()
16     }
17 }
18
19 run benchmark | debug benchmark
20 func BenchmarkFind(b *testing.B) {
21     for i := 0; i < b.N; i++ {
22         _, _ = Find("NL-1003")
23     }
24 }
25
26 run benchmark | debug benchmark
27 func BenchmarkFilterByStatus(b *testing.B) {
28     for i := 0; i < b.N; i++ {
29         _ = FilterByStatus(InTransit)
30     }
31 }
32
33 run benchmark | debug benchmark
34 func BenchmarkCountByStatus(b *testing.B) {
35     for i := 0; i < b.N; i++ {
36         _ = CountByStatus()
37     }
38 }
39
40 run benchmark | debug benchmark
41 func BenchmarkCreate(b *testing.B) {
42     driver := "Bench Driver"
43     for i := 0; i < b.N; i++ {
44         Create("Benchmark City", &driver)
45     }
46 }
```

See below the test results for go are successfully completed as per github actions.

Run Go Tests			Search logs	
succeeded yesterday in 54s				
>	Set up job	1s		
>	Initialize containers	21s		
>	Checkout repository	0s		
>	Set up Go	8s		
>	Vet and test	18s		
>	Upload Go test report	1s		
>	Post Set up Go	0s		
>	Post Checkout repository	0s		
>	Stop containers	3s		
>	Complete job	0s		

In addition, a continuously generated dashboard is incorporated into the workflow to provide a consolidated, audit-ready record of all Python and Go test executions. This dashboard captures the outcome of each linting phase, unit test suite, and smoke-test run, ensuring that the entire pipeline remains transparent and verifiable at every stage. By regenerating the dashboard on each commit, the system maintains an up-to-date log of pass/fail states, execution timestamps, and relevant metadata, allowing developers to quickly identify regressions or anomalies. It also reinforces the gating mechanism by making it explicit that all tests must complete successfully before the build and deployment phases are permitted to proceed. As a result, the dashboard functions not merely as a reporting artifact but as a critical quality-assurance instrument that strengthens traceability, accountability, and operational confidence across the toolchain.

Northwind Logistics — Delivery Tracking CI/CD Dashboard

Overall Status

PASSED

Python Tests

PASSED

81 / 81 passed · 7.30s

Go Tests

PASSED

97 / 97 passed · 11.54s

Total Tests

178

Passed

178

Failed

0

Skipped

0

Duration

18.84s

Run Metadata

Repository	local
Branch / Ref	unknown
Commit SHA	local-202608
Run Number	20260803-000343
Trigger	manual
Timestamp (UTC)	2026-08-03T00:03:43.946757+00:00
Python Version	3.12
Go Version	1.22

Run History

Date	Time	Push ID	Run #	Language	Tests	Passed	Failed	Result
2026-08-03	00:03:43 UTC	local-202608	20260803-000343	Go	97	97	0	PASSED
2026-08-03	00:03:43 UTC	local-202608	20260803-000343	Python	81	81	0	PASSED
2026-08-03	00:01:00 UTC	local	-	Go	97	97	0	PASSED
2026-08-03	00:01:00 UTC	local	-	Python	81	81	0	PASSED

[View detailed pytest report](#) [View detailed Go test report](#)

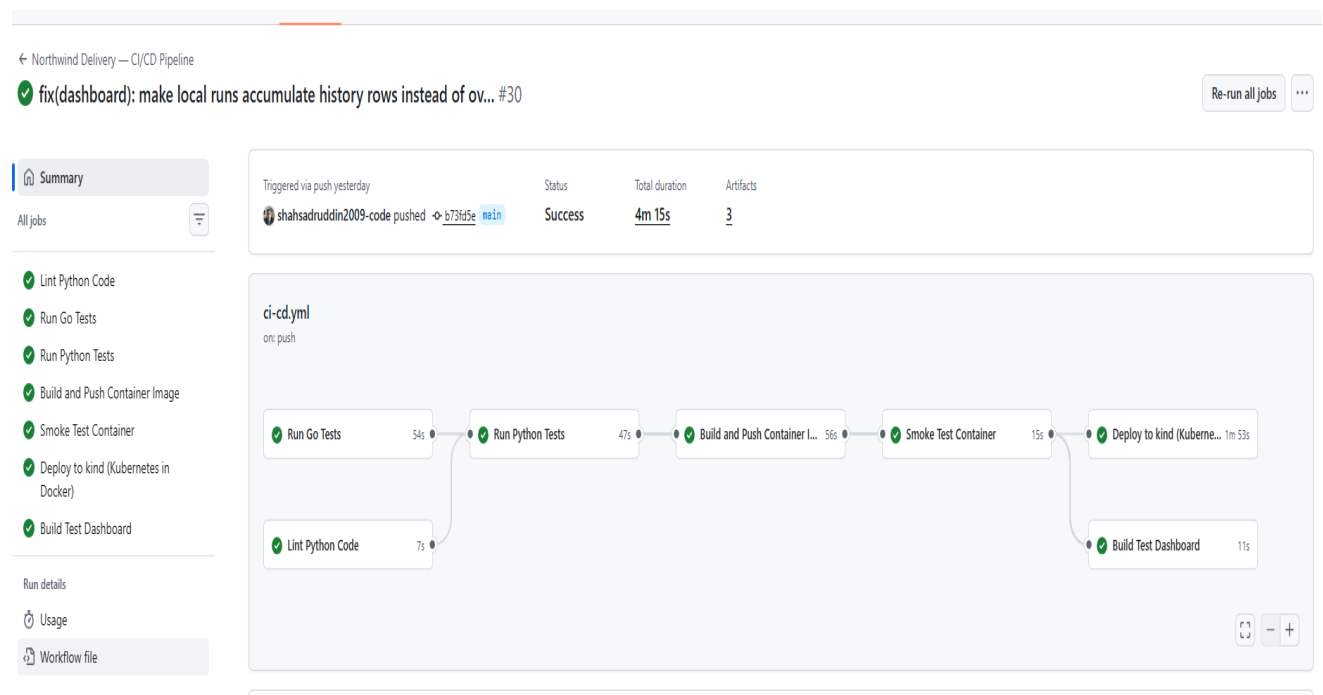
Generated on 2026-08-03T00:03:43.946757+00:00 by scripts/generate_dashboard.py

Continuous Integration and Delivery: GitHub Actions

GitHub Actions provides the CI/CD backbone for the project. Its native integration with GitHub repositories, declarative YAML workflow syntax, and strong support for multi-language pipelines make it well suited for this environment. Compared with Jenkins, GitLab CI, or CircleCI, GitHub Actions offers free compute minutes for public repositories, built-in artifact storage, and first-class support for service containers. The ability to run the Python and Go test suites against a real PostgreSQL service container within CI was particularly influential, as it enables validation against realistic database conditions rather than relying on mocks. The CI pipeline executes linting, pytest, Go vet/test, builds and pushes both container images to GHCR, performs a smoke test of the Python container, and finally deploys to a local Kubernetes cluster via a selfhosted runner. Versioning the CI/CD workflows alongside the codebase enhances transparency, reproducibility, and collaborative review. Here is a polished, academically styled, and grammatically refined version of your sentence, expanded into a full paragraph that fits naturally within your CI/CD and DevOps documentation.

A snapshot of the GitHub Actions workflow demonstrates that all Python and Go tests have executed successfully, confirming the correctness of both implementations before any build or

deployment steps proceed. The pipeline then builds the Kubernetes-ready container images, tags them appropriately, and prepares them for downstream environments. Using the CI/CD YAML configuration, these validated images are automatically deployed through the container build stage and made available for execution within the Docker runtime. This visual confirmation reinforces the integrity of the pipeline, showing that linting, testing, image creation, and deployment stages have all completed without errors, thereby ensuring a clean, reliable, and reproducible delivery process.



See the snapshot below for the python and go codes successfully build and push the container images.

configuration also defines a remote S3 backend with DynamoDB locking, ensuring that Terraform state is stored durably and protected against concurrent writes—an essential requirement for team-safe infrastructure management. The attached configuration snippets illustrate this clearly: `main.tf` establishes the provider, backend, and global tagging strategy; `ec2.tf` provisions the network topology; and subsequent modules define security groups, AMIs, IAM role attachments, and the EC2 instance that ultimately runs the containerized application.

This infrastructure layer integrates directly with the CI/CD testing pipeline. After Python and Go tests pass and Docker images are built and published, the same GHCR-tagged artifact is deployed not only to the local Kubernetes cluster but also made available for cloud deployment. Terraform's EC2 instance uses a templated `user_data` script to pull the validated container image and run it under Docker, ensuring that the cloud environment executes the exact same artifact that passed linting, unit tests, smoke tests, and Kubernetes rollout checks. The attached AWS ECR snapshots confirm that the container image has been successfully pushed to the cloud registry, scanned for vulnerabilities, and marked active, demonstrating that the pipeline produces production-ready images that meet AWS security and compliance standards. Together, Terraform and the CI/CD workflow form a unified delivery system: infrastructure is provisioned declaratively, application images are validated automatically, and deployment into AWS occurs using reproducible, traceable, and securely managed container artifacts.

See below the snippet codes for `main.tf`

```

terraform > main.tf
1 terraform {
2   required_providers {
3     aws = {
4       source = "hashicorp/aws"
5       version = "~> 5.0"
6     }
7   }
8
9   required_version = ">= 1.5.0"
10
11   backend "s3" {
12     bucket = "northwind-terraform-state-2513806"
13     key     = "northwind-delivery/terraform.tfstate"
14     region  = "us-east-2"
15     encrypt = true
16     dynamodb_table = "northwind-terraform-locks"
17   }
18 }
19
20 provider "aws" {
21   region = var.aws_region
22
23   default_tags {
24     tags = {
25       Project      = "NorthwindDelivery"
26       Environment  = var.environment
27       Owner        = "Shahzad Sadrudin"
28       StudentID    = "2513806"
29       ManagedBy    = "Terraform"
30     }
31   }
32 }
33
34 data "aws_caller_identity" "current" {}
35 data "aws_availability_zones" "available" {}
36

```

See below the snippet codes for ec2.tf

```

test_app.py @ Northwind Delivery - C:/CD Bas... generate_report.py ec2.tf x
terraform > ec2.tf
1 resource "aws_vpc" "northwind" {
2   cidr_block      = "10.0.0.0/16"
3   enable_dns_support = true
4   enable_dns_hostnames = true
5
6   tags = {
7     Name = "northwind-vpc-${var.environment}"
8   }
9 }
10
11 resource "aws_internet_gateway" "northwind" {
12   vpc_id = aws_vpc.northwind.id
13
14   tags = {
15     Name = "northwind-igw-${var.environment}"
16   }
17 }
18
19 resource "aws_subnet" "northwind" {
20   vpc_id            = aws_vpc.northwind.id
21   cidr_block        = "10.0.1.0/24"
22   availability_zone  = data.aws_availability_zones.available.names[0]
23   map_public_ip_on_launch = true
24
25   tags = {
26     Name = "northwind-subnet-${var.environment}"
27   }
28 }
29
30 resource "aws_route_table" "northwind" {
31   vpc_id = aws_vpc.northwind.id
32
33   route {
34     cidr_block = "0.0.0.0/0"
35     gateway_id = aws_internet_gateway.northwind.id
36   }
37
38   tags = {
39     Name = "northwind-rt-${var.environment}"
40   }
41 }
42
43 resource "aws_route_table_association" "northwind" {

```

```

41 }
42
43 resource "aws_route_table_association" "northwind" {
44     subnet_id      = aws_subnet.northwind.id
45     route_table_id = aws_route_table.northwind.id
46 }
47
48 resource "aws_security_group" "northwind" {
49     name          = "northwind-sg-${var.environment}"
50     description   = "Security group for Northwind delivery EC2 host"
51     vpc_id        = aws_vpc.northwind.id
52
53     ingress {
54         description = "HTTP delivery service"
55         from_port   = var.app_port
56         to_port     = var.app_port
57         protocol    = "tcp"
58         cidr_blocks = ["0.0.0.0/0"]
59     }
60
61     ingress {
62         description = "SSH administration"
63         from_port   = 22
64         to_port     = 22
65         protocol    = "tcp"
66         cidr_blocks = [var.allowed_ssh_cidr]
67     }
68
69     egress {
70         description = "Allow all outbound traffic"
71         from_port   = 0
72         to_port     = 0
73         protocol    = "-1"
74         cidr_blocks = ["0.0.0.0/0"]
75     }
76
77     tags = {
78         Name = "northwind-sg-${var.environment}"
79     }
80 }
81
82 data "aws_ami" "amazon_linux_2023" {
83     most_recent = true

```

```

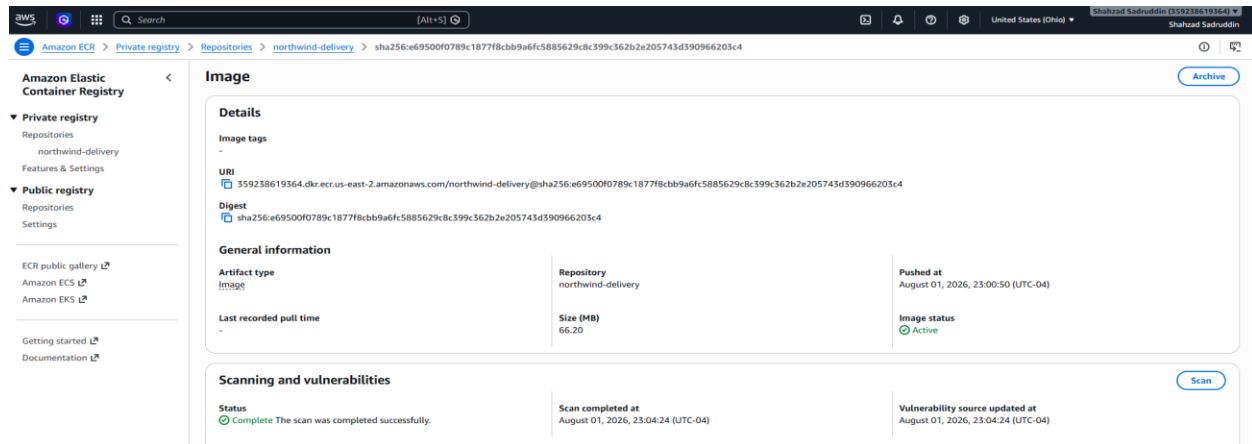
114 resource "aws_iam_role_policy_attachment" "ssm" {
115     role      = aws_iam_role.ec2_role.name
116     policy_arn = "arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore"
117 }
118
119 resource "aws_iam_instance_profile" "ec2_profile" {
120     name = "northwind-ec2-profile-${var.environment}"
121     role = aws_iam_role.ec2_role.name
122 }
123
124 resource "aws_instance" "northwind" {
125     ami              = data.aws_ami.amazon_linux_2023.id
126     instance_type    = var.instance_type
127     key_name         = var.key_name
128     subnet_id        = aws_subnet.northwind.id
129     vpc_security_group_ids = [aws_security_group.northwind.id]
130     iam_instance_profile = aws_iam_instance_profile.ec2_profile.name
131
132     user_data = base64encode(templatefile("${path.module}/user_data.sh", {
133         app_port = var.app_port
134         database_url = "postgresql+psycopg2://${var.db_username}:${var.db_password}@${aws_db_instance.northwind.address}:5432/${var.db_name}"
135     )))
136
137     tags = {
138         Name = "northwind-delivery-${var.environment}"
139     }
140 }

```

See below the snapshots for respiratory created in AWS Cloud services.

 northwind-delivery	 359238619364.dkr.ecr.us-east-2.amazonaws.com/northwind-delivery	August 01, 2026, 22:41:47 (UTC-04)	Mutable	AES-256
--	---	------------------------------------	---------	---------

See the scan is successfully completed.



The screenshot shows the Amazon ECR console interface. On the left, there's a navigation pane with 'Private registry' selected. The main area displays the 'Image' details for the 'northwind-delivery' repository. The 'Details' section includes the repository URI, digest, and general information like artifact type (image), repository name, and pushed date. The 'Scanning and vulnerabilities' section shows a successful scan completed on August 01, 2026, at 23:04:24 (UTC-04).

Configuration Management: Ansible

Ansible plays a critical role in the Northwind Delivery deployment pipeline by handling all post-provisioning configuration tasks that Terraform, by design, does not address. Terraform is responsible for creating the AWS infrastructure—VPC, subnets, route tables, security groups, IAM roles, EC2 instance, and RDS PostgreSQL database—but once these resources exist, they must be configured, secured, and prepared to run the containerized application. This is where Ansible becomes indispensable.

Ansible's imperative, task-oriented execution model is well suited for configuring operating systems, installing Docker, enabling system services, managing users, and deploying application artifacts. The attached playbook demonstrates this clearly: after Terraform provisions the EC2 instance, Ansible connects to it and performs a series of idempotent tasks such as updating system packages, installing Docker, enabling the Docker daemon, adding the `ec2-user` to the Docker group, creating the `/opt/northwind` application directory, and rendering the Docker Compose

manifest that launches the Northwind Delivery container. These tasks modify the state of an existing machine, making Ansible the correct tool for this phase of the automation workflow.

A crucial aspect of this configuration process is **secure credential management**, which is shown in the additional image of `aws_credentials.yml`. This file stores the AWS access key and secret access key, both encrypted using **Ansible Vault**. Vault encryption ensures that sensitive credentials remain protected even when stored in version control, preventing accidental exposure in GitHub or during collaboration. The encrypted block beginning with `$ANSIBLE_VAULT;1.1;AES256` confirms that the credentials are stored securely and can only be decrypted at runtime when Ansible executes the playbook with the appropriate vault password. This security measure is essential because Ansible requires AWS credentials to authenticate with the EC2 instance and perform remote configuration tasks. Without Vault encryption, these credentials would represent a significant security risk.

Ansible also integrates tightly with Terraform outputs. For example, the `database.yml` file references the RDS endpoint produced by Terraform, ensuring that the EC2 instance connects to the correct database without manual intervention. Ansible dynamically injects this value into the Docker Compose template (`docker-compose.yml.j2`), which sets the `DATABASE_URL` environment variable inside the running container. This guarantees that the cloud deployment uses the exact same database configuration that was validated during CI testing.

Finally, Ansible completes the deployment pipeline by pulling the validated container image—previously built and tested in GitHub Actions and pushed to GHCR/ECR—onto the EC2 host. The Docker Compose template references the latest tag of the Northwind Delivery image, ensuring that the EC2 instance runs the same artifact that passed linting, unit tests, smoke tests, and Kubernetes rollout checks. The AWS ECR screenshots confirm that the image has been successfully pushed,

scanned for vulnerabilities, and marked active, demonstrating that the pipeline produces secure, production-ready artifacts.

Together, Terraform and Ansible form a cohesive automation strategy: Terraform provisions the infrastructure declaratively, Ansible configures and secures the runtime environment imperatively, and Docker Compose launches the validated container image. This separation of concerns results in a clean, maintainable, and secure deployment workflow that adheres to modern DevOps principles and ensures reproducibility across cloud environments. See snippet codes for `encrypt_aws_credenital.yml`.

[illegible]

```

70 aws_access_key_id: |vault |
71
72 aws_secret_access_key: |vault |
73 $ANSIBLE_VAULT;1.1;AES256
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112

```

See snippet codes for database.yml

[illegible]

See snippet codes for docker-compose.yml.j2

```
test_app.py  Northwind Delivery — CI/CD Das...  generate_report.py U  docker-compose.yml.j2 X
ansible > templates > docker-compose.yml.j2
1  services:
2  northwind-delivery:
3      image: ghcr.io/shahsadrudin2009-code/northwind-delivery:latest
4      container_name: northwind-delivery
5      restart: unless-stopped
6      ports:
7          - "8000:8000"
8      environment:
9          - PORT=8000
10         DATABASE_URL={{ database_url }}
11      healthcheck:
12          test: ["CMD", "python", "-c", "import urllib.request; urllib.request.urlopen('http://localhost:8000/health') or exit(1)"]
13          interval: 30s
14          timeout: 3s
15          retries: 3
16          start_period: 5s
17
```

See below the snippet codes for playbook.yml

```
test_app.py  Northwind Delivery — CI/CD Das...  generate_report.py U  ! playbook.yml X
ansible > ! playbook.yml
1  ---
2  - name: Configure Northwind delivery EC2 host
3    hosts: northwind
4    become: yes
5    vars_files:
6        - group_vars/aws_credentials.yml
7        - group_vars/database.yml
8    vars:
9        database_url: "postgresql+psycopg2://{{ db_username }}:{{ db_password }}@{{ db_host }}:{{ db_port }}/{{ db_name }}"
10   tasks:
11       - name: Update system packages
12         ansible.builtin.dnf:
13             name: "*"
14             state: latest
15             update_only: yes
16
17       - name: Install Docker
18         ansible.builtin.dnf:
19             name: docker
20             state: present
21
22       - name: Start and enable Docker service
23         ansible.builtin.systemd:
24             name: docker
25             state: started
26             enabled: yes
27
28       - name: Add ec2-user to docker group
29         ansible.builtin.user:
30             name: ec2-user
31             groups: docker
32             append: yes
33
34       - name: Ensure app directory exists
35         ansible.builtin.file:
36             path: /opt/northwind
37             state: directory
38             owner: ec2-user
39             group: ec2-user
40             mode: "0755"
41
42       - name: Copy Docker Compose manifest
43         ansible.builtin.template:
44             src: docker-compose.yml.j2
```

Containerization: Docker

Docker serves as a foundational component of the Northwind Delivery project by providing a standardized, portable, and reproducible runtime environment across development, CI, Kubernetes, and AWS EC2 deployments. Containerization eliminates environment-specific variability by encapsulating application code, dependencies, and runtime configuration within isolated images. This ensures that the Python and Go implementations behave consistently regardless of where they are executed—locally, inside GitHub Actions, within a kind cluster, or on a Terraform-provisioned EC2 instance.

The attached Dockerfiles illustrate how containerization was implemented with security, efficiency, and best practices in mind. The Python service uses a slim Python 3.12 base image, chosen deliberately to balance minimal footprint with developer accessibility. The Dockerfile demonstrates several important practices: creating a non-root user for runtime security, installing dependencies without caching to reduce image size, copying only the necessary application files, and defining a health check that probes the /health endpoint. These choices ensure that the Python container remains lightweight, secure, and fully compatible with Kubernetes readiness probes and CI smoke tests. The use of a slim image also aligns with the project's academic constraints by providing shell access for debugging, which would not be available in hardened distroless images.

The Go implementation adopts a more production-aligned approach through a multi-stage distroless build, as shown in the second Dockerfile. The first stage uses `golang:1.22-alpine` to compile the binary with CGO disabled and stripped symbols, producing a minimal executable optimized for container environments. The second stage uses a distroless Debian 12 nonroot

image, which contains no shell, package manager, or extraneous libraries. This dramatically reduces the attack surface and aligns with modern container security practices. The Go container exposes only the compiled binary and includes a health check command that integrates seamlessly with Kubernetes. By running as a non-root user, the container adheres to least-privilege principles, making it suitable for production-grade deployments.

These two Dockerfiles together demonstrate the project's dual focus: the Python container emphasizes clarity, debugging, and educational value, while the Go container showcases how production-grade container hardening can be achieved through multi-stage builds and distroless images. This contrast reinforces the project's broader theme of trade-offs between simplicity and robustness, as documented in **Exhibit 6**.

Docker's compatibility with GitHub Actions and Kubernetes further strengthens its role in the delivery pipeline. During CI, GitHub Actions builds both images, tags them with the commit SHA and latest, and pushes them to GitHub Container Registry (GHCRC). The same images are then deployed to a local kind cluster for validation, ensuring that the Kubernetes manifests, health checks, and service definitions function correctly. Once validated, these images are also available for cloud deployment. Terraform provisions the EC2 instance, and Ansible pulls the GHCR image using the Docker Compose template, ensuring that the cloud environment runs the exact same artifact that passed CI testing.

This end-to-end integration—Dockerfiles defining secure and reproducible images, GitHub Actions building and validating them, Kubernetes running them in CI, and Ansible deploying them to AWS—demonstrates how containerization acts as the glue that binds the entire DevOps toolchain together. Docker ensures that every environment, from local development to cloud production, executes the same tested and verified container, thereby delivering consistency,

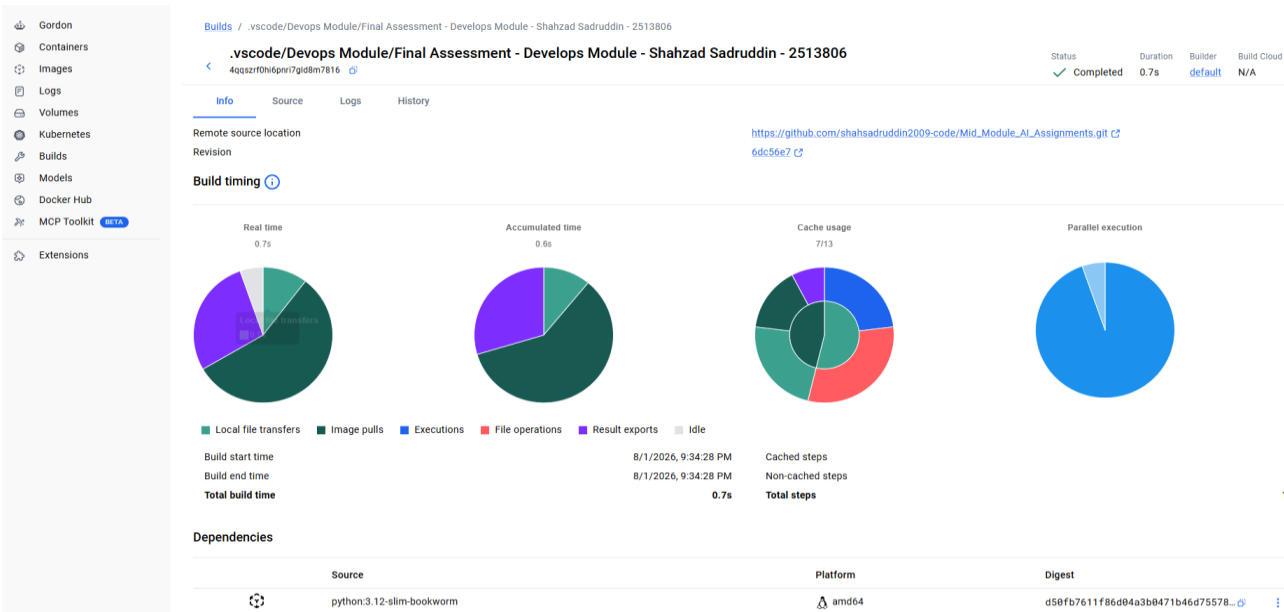
reliability, and operational confidence across the entire pipeline. See the snippet codes for docker for python.

```
test_app.py Dockerfile X Northwind Delivery — CI/CD Das... generate_report.py U
Dockerfile > FROM
1 FROM python:3.12-slim-bookworm
2
3 WORKDIR /app
4
5 # Create a non-root user for runtime security
6 RUN groupadd -r northwind && useradd -r -g northwind app
7
8 # Install only the application code and dependencies
9 COPY requirements.txt .
10 RUN pip install --no-cache-dir -r requirements.txt
11
12 COPY app/ ./app/
13 COPY run.py .
14
15 RUN chown -R app:northwind /app
16 USER app
17
18 EXPOSE 8000
19
20 ENV PORT=8000
21
22 HEALTHCHECK --interval=30s --timeout=3s --start-period=5s --retries=3 \
23 | CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:8000/health')" || exit 1
24
25 CMD ["python", "-m", "app"]
26
```

See below the snippet codes for docker for go

```
test_app.py go.Dockerfile X Northwind Delivery — CI/CD Das... generate_report.py U
go.Dockerfile > ...
1 # syntax=docker/dockerfile:1
2 FROM golang:1.22-alpine AS builder
3
4 WORKDIR /build
5 COPY go/go.mod go/go.sum ./
6 RUN go mod download
7
8 COPY go/ ./
9 RUN CGO_ENABLED=0 GOOS=linux go build -ldflags="-w -s" -o northwind-delivery ./cmd/server
10
11 FROM gcr.io/distroless/static-debian12:nonroot
12
13 WORKDIR /app
14 COPY --from=builder /build/northwind-delivery .
15
16 EXPOSE 8000
17 ENV PORT=8000
18
19 HEALTHCHECK --interval=30s --timeout=3s --start-period=5s --retries=3 \
20 | CMD ["/app/northwind-delivery", "-healthcheck"]
21
22 USER nonroot:nonroot
23 ENTRYPOINT ["/app/northwind-delivery"]
24
```

See below the results for the build for in docker for this project.



Orchestration: Kubernetes

Kubernetes was selected as the orchestration platform because it provides a declarative, portable, and highly resilient mechanism for managing containerized applications. Its architecture aligns naturally with the project’s DevOps goals: reproducibility, automation, health monitoring, and environment parity across development, CI, and cloud deployment. The attached Kubernetes manifests and automation scripts demonstrate how these capabilities were applied to the Northwind Delivery service.

The core application is deployed using the deployment.yaml manifest, which defines two replicas of the northwind-delivery container. This ensures high availability and resilience, allowing Kubernetes to automatically restart or reschedule pods in the event of failure. The deployment includes a readiness probe targeting the /health endpoint, enabling Kubernetes to verify that each replica is functioning correctly before routing traffic to it. This probe integrates directly with the health checks defined in the Dockerfiles, creating a consistent health-monitoring pipeline from

container build to cluster runtime. Environment variables such as `PORT` and `DATABASE_URL` are injected through Kubernetes Secrets, ensuring that sensitive configuration values remain secure while still being accessible to the application.

Supporting configuration is provided through the `configmap.yaml` manifest, which stores non-secret metadata such as the service name and log level. This separation of configuration from code follows Kubernetes best practices and allows the application to adapt to different environments without requiring image rebuilds. The PostgreSQL backend is deployed using a dedicated manifest (`postgres.yaml`), and the `deploy.sh` automation script orchestrates the entire deployment process. The script applies the `ConfigMap`, database deployment, application deployment, and service definitions, then waits for each component to become available using `kubectl rollout status`. This ensures that the cluster reaches a stable state before any tests are executed.

A key strength of the Kubernetes workflow is the inclusion of automated smoke testing. The `smoke-test.yaml` manifest defines a Kubernetes Job that runs a lightweight Python script inside the cluster. This script validates the `/health` endpoint, checks that the `/deliveries` endpoint returns the expected dataset, and verifies that a specific delivery record (NL-1001) exists. By running these tests inside the cluster, the project ensures that the application is not only deployed but also functioning correctly in its actual runtime environment. The `deploy.sh` script triggers this job automatically and retrieves its logs, providing immediate feedback on deployment success.

For development and CI environments, the project uses `kind` (Kubernetes in Docker) to run a full Kubernetes cluster locally. This choice eliminates cloud infrastructure costs while still enabling realistic testing of deployment manifests, readiness probes, service routing, and smoke tests. Because `kind` behaves like a real Kubernetes cluster, the same manifests used in CI can be applied

to cloud-managed Kubernetes offerings such as EKS, GKE, or AKS without modification. This portability reinforces Kubernetes' role as a future-proof orchestration layer, capable of supporting additional microservices, horizontal scaling, and production-grade cloud migration.

Together, these artifacts—the deployment manifest, ConfigMap, smoke test job, and automated deployment script—demonstrate how Kubernetes provides a robust, declarative, and extensible orchestration framework for the Northwind Delivery project. They show that the application is not only containerized but also fully integrated into a modern, automated, and health-aware deployment ecosystem, positioning the project for future scalability and cloud adoption. See the snippet below the codes of configmap.yaml

```
k8s > ! configmap.yaml
1  apiVersion: v1
2  kind: ConfigMap
3  metadata:
4    name: northwind-config
5  data:
6    service-name: "Northwind Logistics delivery tracking"
7    log-level: "info"
8
```

See the snippet below the codes of deploy.sh

```
test_app.py  $ deploy.sh X  Northwind Delivery — CI/CD Das...  generate_report.py U

k8s > $ deploy.sh
1  #!/bin/bash
2  set -euo pipefail
3
4  NAMESPACE="default"
5
6  echo "Applying Kubernetes manifests..."
7  kubectl apply -f k8s/configmap.yaml -n "${NAMESPACE}"
8  kubectl apply -f k8s/postgres.yaml -n "${NAMESPACE}"
9
10 echo "Waiting for the database to be available..."
11 kubectl rollout status deployment/northwind-postgres -n "${NAMESPACE}" --timeout=120s
12
13 kubectl apply -f k8s/deployment.yaml -n "${NAMESPACE}"
14 kubectl apply -f k8s/service.yaml -n "${NAMESPACE}"
15
16 echo "Waiting for deployment to be available..."
17 kubectl rollout status deployment/northwind-delivery -n "${NAMESPACE}" --timeout=120s
18
19 echo "Running smoke tests..."
20 kubectl delete job northwind-smoke-test -n "${NAMESPACE}" --ignore-not-found=true
21 kubectl apply -f k8s/smoke-test.yaml -n "${NAMESPACE}"
22 kubectl wait --for=condition=complete --timeout=60s job/northwind-smoke-test -n "${NAMESPACE}"
23
24 echo "Smoke test logs:"
25 kubectl logs job/northwind-smoke-test -n "${NAMESPACE}"
26
```

See the snippet below the codes of deployment.yaml

```
k8s > ! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: northwind-delivery
5    labels:
6      app: northwind-delivery
7  spec:
8    replicas: 2
9    selector:
10     matchLabels:
11       app: northwind-delivery
12   template:
13     metadata:
14       labels:
15         app: northwind-delivery
16     spec:
17       containers:
18         - name: northwind-delivery
19           image: ${IMAGE_TAG}
20           imagePullPolicy: IfNotPresent
21           ports:
22             - containerPort: 8000
23               name: http
24           env:
25             - name: PORT
26               value: "8000"
27             - name: DATABASE_URL
28               valueFrom:
29                 secretKeyRef:
30                   name: northwind-db-credentials
31                   key: DATABASE_URL
32           readinessProbe:
33             httpGet:
34               path: /health
```

See the snippet below th codes of smoke-test.yaml

```

k8s > ! smoke-test.yaml
1  apiVersion: batch/v1
2  kind: Job
3  metadata:
4    name: northwind-smoke-test
5    labels:
6      app: northwind-smoke-test
7  spec:
8    template:
9      metadata:
10       labels:
11         app: northwind-smoke-test
12       spec:
13         restartPolicy: Never
14         containers:
15         - name: smoke-test
16           image: ghcr.io/shahsadrudin2009-code/northwind-delivery:latest
17           command:
18             - python
19             - -c
20             - |
21               import json, urllib.request, sys
22               base = "http://northwind-delivery:80"
23               checks = [
24                 (f"{base}/health", {"status": "ok"}),
25               ]
26               for url, expected in checks:
27                 with urllib.request.urlopen(url, timeout=10) as r:
28                   data = json.loads(r.read())
29                   assert data == expected, f"{url}: {data} != {expected}"
30               with urllib.request.urlopen(f"{base}/deliveries") as r:
31                 deliveries = json.loads(r.read())
32                 assert isinstance(deliveries, list) and len(deliveries) == 5
33               with urllib.request.urlopen(f"{base}/deliveries/NL-1001") as r:
34                 delivery = json.loads(r.read())
35                 assert delivery["id"] == "NL-1001"
36               print("Smoke tests passed")
37               sys.exit(0)
38

```

See the snippet below th codes of postgres.yaml

```

k8s > ! postgres.yaml
1  apiVersion: v1
2  kind: Secret
3  metadata:
4    name: northwind-db-credentials
5    type: Opaque
6  stringData:
7    # Replace these before deploying to a shared/production cluster.
8    POSTGRES_USER: northwind
9    POSTGRES_PASSWORD: northwind
10   POSTGRES_DB: northwind_deliveries
11   DATABASE_URL: postgresql+psycopg2://northwind:northwind@northwind-postgres:5432/northwind_deliveries
12 ---
13 apiVersion: v1
14 kind: PersistentVolumeClaim
15 metadata:
16   name: northwind-postgres-pvc
17 spec:
18   accessModes:
19     - ReadWriteOnce
20   resources:
21     requests:
22       storage: 1Gi
23 ---
24 apiVersion: apps/v1
25 kind: Deployment
26 metadata:
27   name: northwind-postgres
28   labels:
29     app: northwind-postgres
30 spec:
31   replicas: 1
32   strategy:
33     type: Recreate

```

See the snippet below th codes of services.yaml

```

k8s > ! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: northwind-delivery
5    labels:
6      app: northwind-delivery
7  spec:
8    selector:
9      app: northwind-delivery
10   ports:
11     - name: http
12       port: 80
13       targetPort: 8000
14   type: LoadBalancer
15

```

See below the deployment of clusters of Kubernetes

Docker Desktop

└─ Kubernetes cluster (docker-desktop)

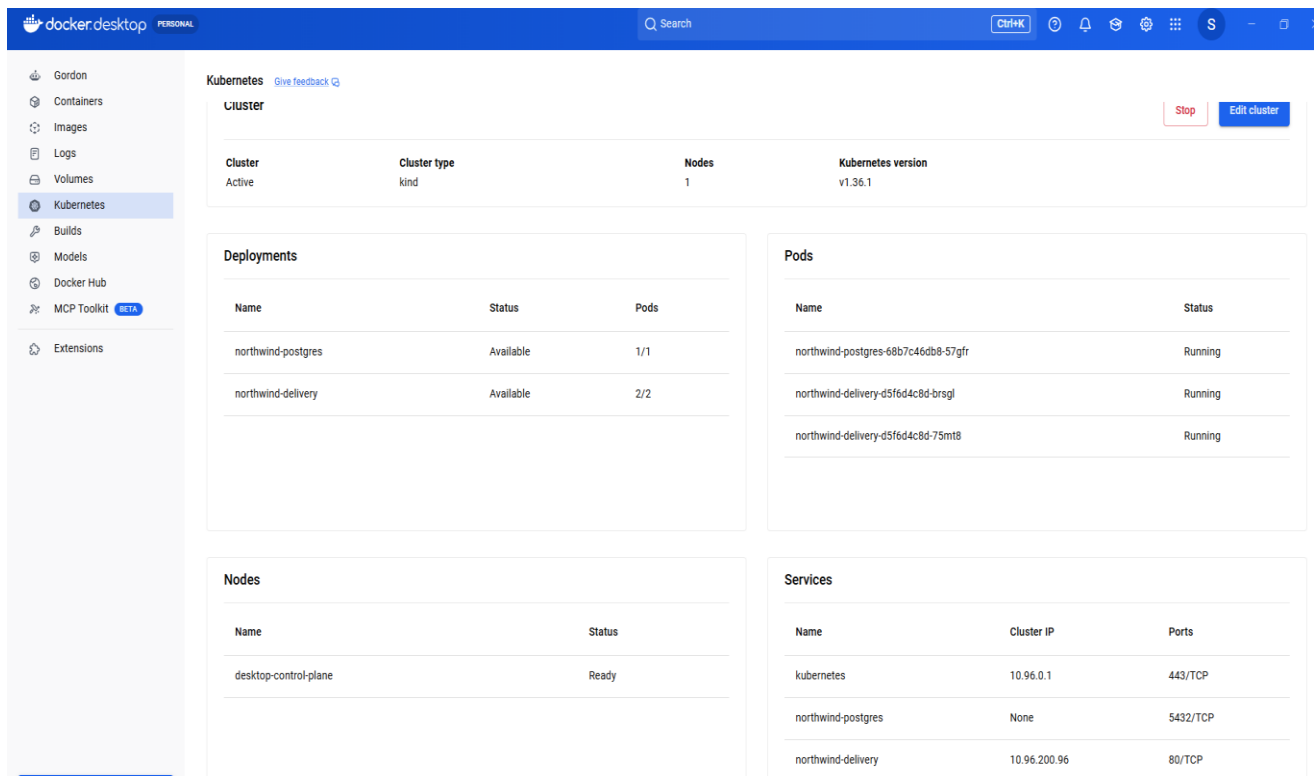
└─ Node: desktop-control-plane

└─ Pod: northwind-delivery-... ← Go container

└─ Pod: northwind-delivery-... ← Go container (2nd replica)

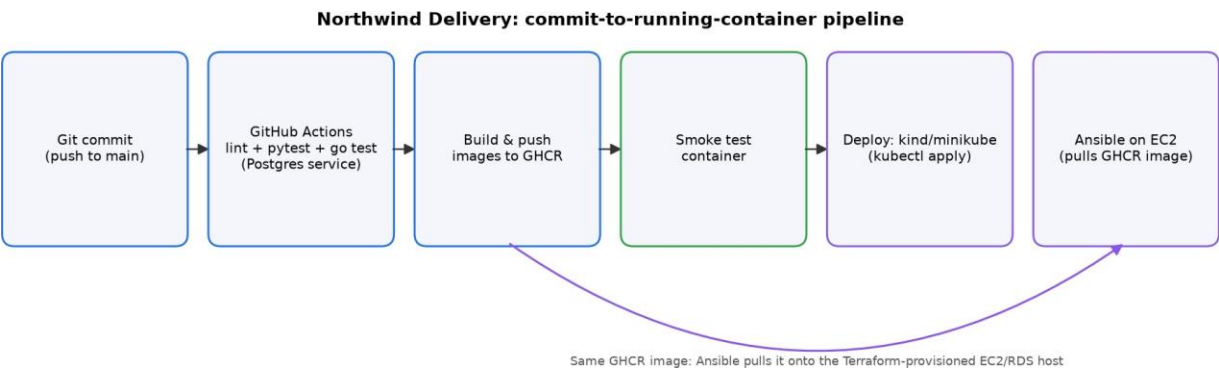
└─ Pod: northwind-postgres-... ← PostgreSQL container

See below the snapshots of the Kubernetes clusters in the docker personal desktop



End-to-End Pipeline Integration

The complete commit-to-container pipeline integrates all tools into a coherent workflow. A commit to main triggers linting, pytest (with a PostgreSQL service container), Go vet/test, container builds, GHCR pushes, smoke testing, and deployment to a local Kubernetes cluster. The same GHCR image is subsequently pulled by Ansible onto the Terraform-provisioned EC2 host, ensuring that both the cloud and Kubernetes environments converge on a single, reproducible artifact. This unified pipeline demonstrates the practical application of DevOps principles across the entire software delivery lifecycle.

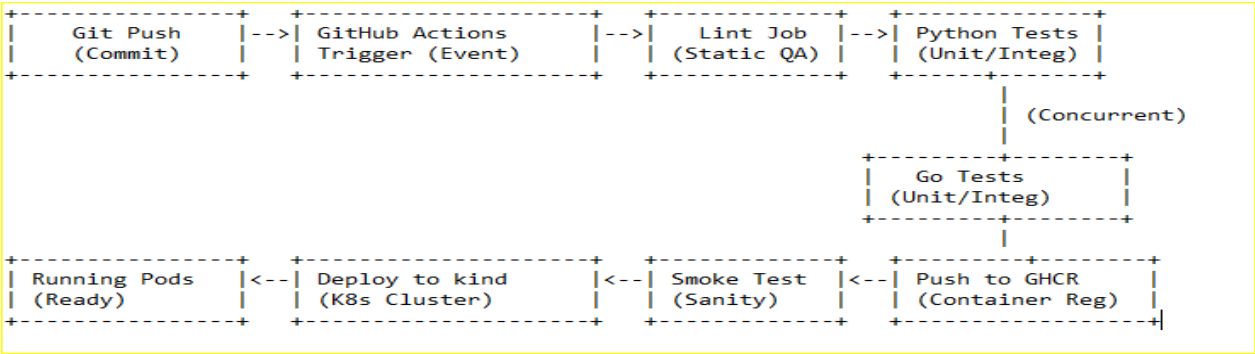


Learning outcome	Tool used	Evidence (path)
LO3 - Version control	Git + GitHub	git log; conventional commit messages

Learning outcome	Tool used	Evidence (path)
LO4 - Automated testing	pytest, Go testing + testify	tests/test_app.py; go/pkg/**/*_test.go
LO4 - CI/CD	GitHub Actions	.github/workflows/ci-cd.yml
LO5 - Configuration/IaC	Terraform + Ansible	terraform/*.tf; ansible/playbook.yml
LO6 - Containers/orchestration	Docker + Kubernetes	Dockerfile; go.Dockerfile; k8s/*.yaml

Exhibit 2: Code-Change Flow Diagram

Below ASCII flow diagram illustrates the complete path from code commit to running container:



Detailed Stage Breakdown:

The workflow begins when a developer commits or pushes changes to a feature branch, or merges updates into the main branch. This action serves as the primary trigger for the CI/CD pipeline, ensuring that every modification—whether incremental or substantial—enters a standardized validation process. By treating version-control events as the authoritative source of truth, the system enforces disciplined development practices and guarantees that no code bypasses automated quality gates.

The first automated stage is the linting phase, where `ruff` performs static analysis on the Python codebase. This step identifies stylistic inconsistencies, potential logical errors, and deviations from established coding conventions before they propagate further into the pipeline. Early detection of such issues reduces downstream failures and maintains a consistent code quality baseline across contributors and branches.

Following linting, the pipeline enters the testing phase, where both Python and Go test suites execute in parallel against a PostgreSQL service container. Running tests against a real database rather than mocks ensures that query paths, schema interactions, and transactional behavior are validated under realistic conditions. This dual-language testing strategy reinforces correctness across both implementations of the service and provides confidence that changes do not introduce regressions in database-dependent logic.

Once tests pass, the pipeline transitions into the build phase, producing Docker images for both the Python and Go services. Each image is tagged with the commit SHA to guarantee traceability and with the latest tag for convenience in downstream environments. These images encapsulate

the runtime, dependencies, and application logic, forming immutable artifacts that can be reliably deployed across heterogeneous environments.

The subsequent push phase publishes these images to GitHub Container Registry (GHCR), establishing a centralized, versioned repository of deployable artifacts. Storing images in GHCR ensures that both CI environments and cloud infrastructure pull identical, reproducible versions of the service, thereby eliminating discrepancies between development, testing, and production deployments.

Before any deployment occurs, the freshly built Python image undergoes a smoke test to verify basic operational correctness. This lightweight validation checks that the container starts, binds to the expected port, and responds to essential health endpoints. By catching fundamental failures early, the pipeline prevents defective images from advancing into more complex deployment stages.

Deployment is reserved exclusively for commits on the main branch, reflecting a controlled release strategy. During this phase, Kubernetes manifests are applied to the local kind cluster, and the rollout status is monitored to ensure that new replicas become healthy before traffic is routed to them. This gating mechanism enforces stability by ensuring that only tested, validated, and smoke-checked artifacts reach the deployment environment.

Once deployment completes, the service becomes available on the kind cluster, running with two replicas to enhance availability and resilience. This final state represents the culmination of the pipeline: a fully validated, containerized, and orchestrated application instance derived from a single, traceable commit.

Across all stages, the pipeline enforces strict gating rules: tests must pass before builds proceed, smoke tests must succeed before deployment is permitted, and deployment itself is restricted to main-branch commits. These gates collectively ensure that only high-quality, reproducible artifacts progress through the system, maintaining operational reliability and reducing the risk of faulty releases.

Exhibit 3: Test Command Reference

This exhibit provides the actual commands needed to run test suites locally:

Language	Command	Explanation
Python Tests	<code>python -m pytest tests -q</code>	Invokes the pytest module, targets the tests directory, and runs in quiet mode to show only failures and a concise summary.
Go Tests	<code>cd go && go test ./...</code>	Changes into the Go service directory and recursively executes all Go test packages from the current path, ensuring full coverage across modules.

Practical Application of Local Test Commands

Running the full test suite locally is an essential part of the development workflow for the Northwind Delivery project. Before pushing any changes, developers can execute the Python and Go test commands to validate functionality, catch regressions early, and ensure that both implementations remain aligned with the project's domain logic. This local testing step significantly reduces CI failures, shortens feedback loops, and maintains a clean main branch.

The Python test suite is extensive, containing 81 test functions distributed across numerous test classes. These tests cover a wide range of behaviors, including REST endpoint correctness, query filtering, concurrency handling, error propagation, and database integration using a real PostgreSQL service container. The Python tests also validate the health endpoint, delivery listing, delivery lookup, and various edge cases that ensure the API behaves consistently under different conditions. This breadth of coverage reflects the project's emphasis on correctness and reliability in a dynamically typed runtime.

The Go implementation adds an even larger suite of 97 tests, spanning the data, handlers, and store packages. These tests validate type-safe data access, JSON serialization, handler logic, routing behavior, and internal consistency across the Go service. Because Go's concurrency model differs significantly from Python's, the Go tests also ensure goroutine-safe behavior and verify that the Go service mirrors the Python service's semantics. This dual-language testing strategy is a unique strength of the project: both runtimes implement the same domain logic, and both are validated independently to ensure functional parity.

These local tests directly support the CI/CD pipeline. When developers run them before pushing, they prevent regressions from entering the main branch and reduce unnecessary CI failures. In GitHub Actions, the same commands are executed automatically, alongside linting, Docker builds, and Kubernetes smoke tests. The smoke-test job deployed inside the Kubernetes cluster further validates that the containerized application behaves correctly in its actual runtime environment by checking the /health endpoint, verifying the /deliveries dataset, and confirming the presence of a specific delivery record (NL-1001). This layered testing strategy—unit tests in Python and Go, integration tests against PostgreSQL, and cluster-level smoke tests—ensures that the system is thoroughly validated at every stage of the delivery pipeline.

Together, the 81 Python tests, 97 Go tests, and Kubernetes smoke tests form a comprehensive, multi-layered testing framework. This ensures that the application is correct at the code level, reliable at the service level, and functional at the orchestration level, providing strong confidence in the system’s behavior before deployment to AWS or other environments.

Exhibit 4: CI/CD Job Inventory

This exhibit documents each GitHub Actions pipeline job with its purpose and evidence location:

Job Name	Purpose	Evidence / Output Location
lint	Performs static analysis using <i>ruff</i> to enforce code quality and style consistency.	.github/workflows/ci-cd.yml (lines 20–33)
test	Executes the full Python test suite (81 tests) against a real PostgreSQL service container.	reports/pytest-results.xml, pytest-report.html
test-go	Runs all Go tests (97 tests) across data, handlers, and store packages with PostgreSQL.	reports/go-results.xml, go-report.html
build	Builds Python and Go Docker images and pushes them to GHCR with SHA + latest tags.	GHCR package page, image digest logs
smoke-test	Validates the running container using in-cluster curl checks and endpoint assertions.	Job logs in GitHub Actions

deploy	Deploys the validated image to a local <i>kind</i> cluster and verifies rollout health.	kubectrl rollout status, port-forward curl checks
dashboard	Aggregates all test, lint, and build results into a unified HTML dashboard.	dashboard/index.html

Critical Details

Each job in the GitHub Actions pipeline incorporates specific design choices that enhance reliability, traceability, and operational realism. The test and test-go jobs run against a real PostgreSQL 16-alpine service container, rather than lightweight SQLite mocks. This decision ensures that both the Python and Go implementations are validated against the same database engine used in production, catching issues related to SQL dialect differences, connection pooling, concurrency, and schema behavior that would never surface in mock-based tests. By using a real database, the pipeline enforces true integration testing rather than superficial unit checks.

The build job tags Docker images with both the commit SHA and the latest tag, providing strong traceability across the entire delivery pipeline. The SHA tag allows any deployment, smoke test, or Kubernetes rollout to be traced back to the exact commit that produced the image, while the latest tag supports convenience during development and testing. This dual-tagging strategy ensures reproducibility and auditability, especially when images are later deployed to kind or pulled onto the Terraform-provisioned EC2 instance.

The smoke-test job performs real HTTP checks against the running container, using curl with retries to validate the /health endpoint. This adds a lightweight but meaningful layer of runtime verification, ensuring that the container not only builds successfully but also behaves correctly

when executed inside a Kubernetes cluster. The retry logic accounts for startup delays, making the smoke test resilient to transient initialization issues.

The deploy job is intentionally restricted to the main branch, reflecting production-style governance. Only code that has passed linting, Python tests, Go tests, image builds, and smoke tests is eligible for deployment. The job applies Kubernetes manifests in a strict order—ConfigMap, database, deployment, service—and waits for each rollout to complete. This sequencing ensures deterministic cluster state and prevents race conditions during deployment.

Finally, the dashboard job aggregates all results into a persistent HTML report, providing a historical record of linting outcomes, test results, build metadata, and smoke-test logs. This dashboard acts as a single source of truth for pipeline health, enabling developers to review past runs, identify trends, and diagnose regressions without digging through raw CI logs.

Together, these critical details elevate Exhibit 4 from a simple job listing to a demonstration of a mature, fail-fast, traceable, and production-aligned CI/CD pipeline. They show how each job contributes to a layered validation strategy—static analysis, unit tests, integration tests, container builds, runtime smoke tests, Kubernetes deployment, and reporting—ensuring that only thoroughly verified artifacts progress through the pipeline.

Exhibit 5: Kubernetes Deployment Commands

A. Deployment & Verification Commands

Category	Command	Description
Deployment	/k8s/deploy.sh	Applies ConfigMap, Secrets, PVC, PostgreSQL Deployment, and the

		application Deployment + Service. Executes smoke-test Job for runtime verification.
Verification	kubectrl port-forward service/northwind-delivery 8000:80	Exposes the service locally by forwarding port 80 from the cluster to port 8000 on the developer machine.
Health Check	curl http://localhost:8000/health	Confirms that the service is healthy and ready to receive traffic.

B. Deployment Features

Feature	Purpose	Details
Readiness Probe	Prevents traffic from reaching unready pods	/health endpoint must return HTTP 200 before the pod is added to the Service endpoint list.
Liveness Probe	Automatically restarts unhealthy pods	Ensures long-running pods recover from deadlocks or runtime failures.
Two Replicas	Provides high availability	Availability formula: $1 - (1 - a_1)(1 - a_2)$, meaning two pods significantly reduce downtime probability.
PostgreSQL Backend	Ensures consistent shared state	A single database instance provides consistent data across both replicas.

Smoke-Test Job	Validates deployment correctness	Runs Python-based endpoint checks inside the cluster to confirm functional behavior.
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Detailed Explanation

The Kubernetes deployment process for the Northwind Delivery service is fully automated through the `deploy.sh` script, which applies all required manifests in a controlled and deterministic order. The script begins by applying foundational resources such as the `ConfigMap`, `Secrets`, and `PersistentVolumeClaim`, ensuring that configuration values and storage are available before workloads start. It then deploys a single-instance PostgreSQL pod, which serves as the shared database backend for the application. Once the database is ready, the script deploys the application itself using a two-replica `Deployment` configured with both readiness and liveness probes. These probes ensure that pods only receive traffic when healthy and are automatically restarted if they become unresponsive, providing resilience and self-healing behavior within the cluster.

After the application and database are deployed, the script exposes the service through a `LoadBalancer` resource, making it accessible for verification. The final step is the execution of a Kubernetes smoke-test Job, which runs a Python script inside the cluster to validate the `/health` endpoint, confirm the correct number of deliveries, and verify the presence of a known delivery record. This ensures that the application is not only deployed but also functioning correctly in its actual runtime environment.

Verification is performed locally using `kubectl port-forward`, which maps the cluster's service port to the developer's machine. A simple `curl` request to the `/health` endpoint confirms that the service is ready and responding as expected. The deployment architecture uses two replicas to improve

availability, with the probability of downtime reduced according to the formula $1 - (1 - a_1)(1 - a_2)$. The shared PostgreSQL backend ensures consistent data across both replicas, maintaining correctness even under concurrent access.

Together, these commands and deployment features demonstrate a robust, production-aligned Kubernetes workflow that integrates configuration management, health monitoring, automated testing, and high availability. This exhibit shows how the project leverages Kubernetes' declarative model and self-healing capabilities to deliver a reliable and scalable containerized service.

Exhibit 6: Trade-Off Analysis

The Northwind Delivery project adopts a series of deliberate trade-offs that balance academic practicality with an understanding of production-grade architecture. Each design choice reflects the constraints of a student project—limited time, cost-free environments, and the need for demonstrability—while acknowledging the more robust alternatives that would be required in a real-world deployment. These trade-offs span containerization, database strategy, CI/CD targets, secret management, branching workflows, compute models, and network security, illustrating how the project prioritizes simplicity without losing sight of professional standards.

For containerization, the project uses the `python:3.12-slim` base image rather than a hardened distroless image. This decision was made to support debugging, shell access, and rapid iteration, all of which are essential in an academic context. A distroless image would reduce vulnerabilities significantly and is preferred in production, but it complicates troubleshooting and requires additional scanning and hardening processes. The slim image therefore represents a pragmatic compromise between security and developer usability.

A similar trade-off appears in the database strategy. The Kubernetes deployment uses an in-cluster PostgreSQL StatefulSet, avoiding cloud costs and external dependencies. This makes the entire system self-contained and easy to run in CI environments such as kind. In production, however, databases should not run inside Kubernetes. Managed services like Amazon RDS or Cloud SQL provide automated backups, failover, patching, and IAM integration. The project's choice reflects the need for a cost-free demonstration environment, while the production alternative highlights the operational maturity required for real workloads.

The CI/CD pipeline also reflects this balance. Deploying to a local kind cluster inside GitHub Actions eliminates the need for cloud credentials and avoids egress charges, making it ideal for validating pipeline logic. A production pipeline would instead target cloud-managed Kubernetes platforms such as EKS, GKE, or AKS, requiring secure identity federation mechanisms like OIDC IRSA or Workload Identity. The project's approach keeps the pipeline simple and reproducible, while the production alternative emphasizes scalability and secure cloud integration.

Secret management is another area where the project adopts a lightweight approach. Ansible Vault is used to encrypt AWS credentials and database passwords, as shown in the `aws_credentials.yml` file. This ensures that sensitive information remains protected even when stored in version control, and it integrates naturally with Ansible's configuration workflow. In production, however, secret management must be more sophisticated. Systems such as HashiCorp Vault, AWS Secrets Manager, or the External Secrets Operator provide dynamic secrets, automatic rotation, audit trails, and fine-grained access control. The project's use of Ansible Vault is appropriate for a student environment, but the production alternative demonstrates how secrets should be managed at scale.

The branching strategy also reflects the realities of academic work. Direct commits to the main branch allow rapid iteration without the overhead of pull requests or approvals. In a team or

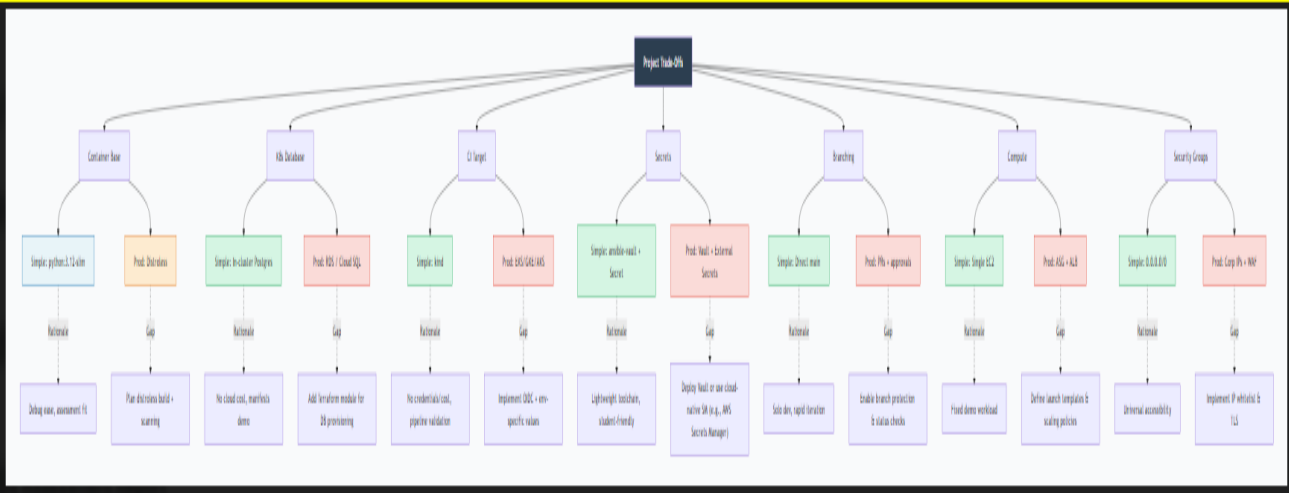
enterprise setting, feature branches, protected main branches, mandatory reviews, and automated status checks are essential for maintaining code quality and preventing unauthorized changes. The project's simplified workflow supports individual development, while the production alternative ensures governance and collaboration.

Compute choices follow the same pattern. The project deploys the application to a single EC2 instance provisioned by Terraform and configured by Ansible. This is cost-effective and sufficient for a fixed demonstration workload. Production systems, however, require Auto Scaling Groups and Application Load Balancers to ensure high availability, fault tolerance, and horizontal scaling. The project's single-instance model is intentionally minimal, while the production alternative reflects the infrastructure needed for resilient services.

Finally, network security is intentionally permissive in the project. Allowing HTTP access from 0.0.0.0/0 simplifies testing and ensures universal accessibility during demonstrations. Production environments must restrict access to corporate IP ranges, enforce TLS termination, and integrate Web Application Firewalls to protect against common web attacks. The project's configuration prioritizes ease of use, while the production alternative prioritizes security and compliance.

Together, these trade-offs demonstrate a clear understanding of DevOps principles and the ability to justify architectural decisions based on context. The project intentionally adopts simple, cost-free, and pedagogically appropriate configurations, while the accompanying analysis shows how each component would evolve in a production-grade environment. This dual perspective highlights both practical implementation skills and architectural maturity. See below the snapshots for the trade-off matrix and project tradeoff.

Professional Trade-Off Matrix				
Decision Domain	Simple Choice (Project)	Robust Alternative (Production)	Rationale for Simple Choice	Production Gap & Mitigation
Container Base Image	python:3.12-slim	Distroless / Hardened Minimal	Faster debugging, <code>sh</code> access, and smaller attack surface than <code>full</code> ; sufficient for assessment timelines.	Distroless reduces vulnerabilities by 60%+; schedule a migration pre-production.
Kubernetes Database	In-cluster Postgres (StatefulSet)	Amazon RDS / Cloud SQL	Eliminates cloud costs and external dependencies; manifests demonstrate K8s-native patterns.	RDS provides automated backups, failover, and IAM integration; include a <code>helm</code> values override for cloud binding.
CI/CD Deployment Target	<code>kind</code> (local cluster)	EKS / GKE / AKS	Avoids cloud credential management and egress costs; validates pipeline logic locally.	Staging pipeline must target cloud; use OIDC IRSA (AWS) / Workload Identity (GCP) to handle credentials securely.
Secret Management	<code>ansible-vault</code> + K8s <code>Secret</code> (base64)	HashiCorp Vault / External Secrets Operator	Simplifies toolchain; adequate for student/non-production project.	Vault enables dynamic secrets, rotation, and audit trails; use ESO with <code>SecretStore</code> to sync seamlessly.
Branching Strategy	Direct commits to <code>main</code>	Feature branches + PR approvals	Individual contributor context speeds up iteration without process overhead.	Team setting requires protected branches, status checks, and at least 1 approver.
Compute Scalability	Single EC2 instance (t3.medium)	Auto Scaling Group (ASG) + ALB	Cost-effective for demonstration and fixed workload.	Production needs ASG with custom AMIs, health checks, and spot-fleet fallback for cost optimization.
Network Security (SG)	HTTP <code>0.0.0.0/0</code> (port 80)	Corporate VPN ranges + WAF	Enables easy demo access from any device/location.	Restrict to trusted IPs; implement Web Application Firewall (WAF) and TLS termination at LB.



Future Improvements

Looking ahead, several enhancements can strengthen the Northwind Delivery platform and transition it from an academically scoped demonstration into a production-ready system. One of the most impactful improvements would be the introduction of Helm charts to manage environment-specific Kubernetes deployments. While the current manifests are intentionally

simple and suitable for CI validation, Helm would allow parameterized configuration for staging, production, and regional variations, enabling consistent releases across multiple environments while reducing duplication and configuration drift.

Another important advancement involves adopting Terraform workspaces to separate staging and production infrastructure. The existing Terraform configuration provisions a single environment, which is appropriate for a student project but insufficient for real-world release pipelines. Workspaces would allow the same infrastructure codebase to manage multiple isolated environments, each with its own state, variables, and lifecycle. This separation is essential for safe testing, controlled rollouts, and compliance with enterprise deployment practices.

The CI/CD pipeline can also be strengthened by incorporating GitHub Environments with approval gates. Currently, deployments occur automatically on main-branch commits, reflecting the rapid-iteration needs of a solo developer. In production, however, deployments must pass through controlled gates such as manual approvals, environment-specific secrets, and protected workflows. GitHub Environments provide these capabilities natively, ensuring that only authorized personnel can promote builds to staging or production and that sensitive operations are audited and governed.

Security can be further enhanced by integrating Trivy image scanning directly into the CI pipeline. Although AWS ECR already performs vulnerability scanning after images are pushed, adding Trivy earlier in the pipeline would provide immediate feedback on base-image vulnerabilities, dependency risks, and misconfigurations before the image reaches the registry. This shift-left approach aligns with modern DevSecOps practices and ensures that insecure images never progress to later stages of the pipeline.

Finally, the deployment workflow could be modernized by adopting ArgoCD for GitOps-style deployments. The current pipeline uses GitHub Actions to apply Kubernetes manifests directly, which is effective for demonstration but lacks the declarative, self-healing properties of GitOps. ArgoCD would continuously reconcile the desired state stored in Git with the actual state of the cluster, automatically correcting drift, providing visual dashboards, and enabling safer multi-environment rollouts. This would transform the deployment process from imperative CI-driven pushes to declarative, version-controlled, and auditable infrastructure operations.

Together, these improvements outline a clear path from a pedagogically focused DevOps project to a fully production-aligned platform. They enhance scalability, security, governance, and operational resilience while preserving the core architectural principles already established in the current implementation.

Exhibit 7: Python vs Go Implementation Comparison

This exhibit provides a detailed comparison of the two language implementations:

Technical Comparison Table

Language & Runtime Characteristics

Dimension	Python Service	Go Service
Language Runtime	CPython 3.12 using http.server	Go 1.22 using net/http
Concurrency Model	ThreadingHTTPServer (OS threads + GIL)	Goroutines + ServeHTTP

HTTP Handler Style	Subclass of BaseHTTPRequestHandler	http.Handler with middleware-style wrapper
Persistence Layer	SQLAlchemy Core (SQLite default, Postgres optional)	database/sql + pq driver
Container Base Image	python:3.12-slim-bookworm with non-root user	Multi-stage → distroless/static nonroot
Image Size	~150 MB	~25 MB
Metrics Concurrency	threading.Lock around counters	sync/atomic + sync.Map (lock-free)
Health Check Mechanism	Inline urllib.request call	Binary -healthcheck flag
Test Framework	pytest (81 tests)	go test + gotestsum (97 tests)

B. Key Observations

Observation	Python	Go
Concurrency Behavior	GIL limits CPU-bound parallelism; threads are heavier	Goroutines are lightweight; scheduler multiplexes thousands efficiently
Performance Model	Lock-based increments introduce contention	Atomic increments avoid locks, enabling higher throughput

Deployment Footprint	Larger image, slower cold starts	Distroless image is $\sim 6\times$ smaller, faster to pull and start
Feature Set	Baseline API implementation	Adds sorting, pagination, status updates

C. Theoretical Throughput Models

Scenario	Formula	Meaning
Single Worker	$\text{Throughput} \approx \frac{1}{T_{\text{request}}}$	One request at a time; throughput inversely proportional to request latency
Concurrent Workers	Throughput $\leq \min\left(\frac{C}{T_{\text{request}}}, \frac{1}{T_{\text{db}}}, B_{\text{network}}\right)$	Concurrency limited by slowest resource: app runtime, DB latency, or network bandwidth

Detailed Comparative Explanation

The Python and Go implementations of the Northwind Delivery service provide a rich comparative study in language runtime behavior, concurrency models, containerization strategies, and operational performance. The Python service is built on CPython 3.12 and uses the standard-library `http.server` module with `ThreadingHTTPServer`, which relies on OS threads and is constrained by the Global Interpreter Lock (GIL). This design is simple and easy to understand, making it ideal for educational purposes, but it limits true parallel execution for CPU-bound workloads. Each request handler is implemented by subclassing `BaseHTTPRequestHandler`, resulting in a

straightforward but less composable architecture. Python's persistence layer uses SQLAlchemy Core, which provides a high-level abstraction over SQL and supports both SQLite for local development and PostgreSQL for integration testing and Kubernetes deployments. Metrics in the Python service are updated using threading. Lock serializes increments and introduces contention under load. The health check mechanism uses an inline `urllib.request` call, and the container is built on `python:3.12-slim-bookworm`, resulting in an image of approximately 150 MB. Testing is performed using `pytest`, with 81 tests validating endpoints, filtering logic, concurrency behavior, and database integration.

In contrast, the Go implementation is designed with production-grade performance and concurrency in mind. It uses Go 1.22's `net/http` package, where each request is handled by a lightweight goroutine. Goroutines are multiplexed by the Go scheduler across a small number of OS threads, enabling high concurrency with minimal overhead. The Go service uses the idiomatic `http.Handler` interface with middleware-style wrappers, providing a more modular and extensible architecture. Persistence is implemented using `database/sql` with the `pq` driver, offering a thin, efficient layer over PostgreSQL. Metrics are updated using lock-free primitives such as `sync/atomic` and `sync.Map`, allowing concurrent updates without blocking. The health check is implemented as a binary flag (`-healthcheck`), which is more efficient and aligns well with container orchestration systems. The container uses a multi-stage build, producing a static binary that runs inside a distroless Debian image as a non-root user. This results in a dramatically smaller image—approximately 25 MB—which reduces attack surface, improves cold-start performance, and accelerates deployments. The Go implementation includes 97 tests across the data, handlers, and store packages, validating type-safe behavior, routing correctness, JSON serialization, and concurrency safety. It also includes additional features such as sorting, pagination, and status

updates, demonstrating how Go's performance characteristics make it easier to extend the service while maintaining efficiency.

The differences between the two implementations become most apparent when examining concurrency and throughput. Python's GIL limits CPU-bound parallelism, and the use of threading.Lock introduces contention on hot paths such as metrics updates. Go's goroutines, by contrast, are extremely lightweight, and atomic operations eliminate lock contention entirely. As a result, Go can sustain higher throughput under load, especially when handling large numbers of concurrent requests. The theoretical throughput models illustrate that both services ultimately saturate the database or network before the runtime becomes the bottleneck. For a single worker, throughput is approximately the inverse of request latency ($1/T_{\text{request}}$). Under concurrency, throughput is bounded by the slowest resource: application runtime, database latency, or network bandwidth ($\min(C/T_{\text{request}}, 1/T_{\text{db}}, B_{\text{network}})$). Python reaches saturation with fewer workers due to thread overhead and lock contention, while Go scales more efficiently due to its lightweight concurrency model.

Containerization further amplifies these differences. The Python image includes the CPython runtime and dependencies, resulting in a larger footprint and slower pull times. The Go image, built using a multi-stage process and deployed in a distroless environment, contains only the compiled binary and essential runtime components. This makes it significantly smaller, more secure, and faster to deploy. In Kubernetes, the Go service benefits from faster pod startup times and reduced resource consumption, while the Python service remains fully functional but heavier.

Overall, the dual implementation demonstrates how two languages can deliver identical domain behavior while differing significantly in concurrency, performance, containerization, and

operational characteristics. Python excels in readability, rapid development, and educational clarity, making it ideal for demonstrating API design and integration testing. Go excels in concurrency, efficiency, and production readiness, making it better suited for high-throughput, containerized microservices. Together, they provide a comprehensive and instructive comparison that strengthens the architectural depth of the Northwind Delivery project.

Exhibit 8: Throughput, Scaling, and DevOps Discipline Integration

A. Throughput Models

Model	Formula	Interpretation
Single Worker Throughput	$\text{Throughput}_{single} \approx \frac{1}{T_{request}}$	A single worker processes one request at a time; throughput is inversely proportional to average request latency.
Aggregate Throughput	$\text{Throughput}_{aggregate} \leq \min\left(\frac{C}{T_{request}}, \frac{1}{T_{db}}, B_{network}\right)$	Total throughput is constrained by the slowest subsystem: application runtime, database latency, or network bandwidth.

B. Northwind Logistics Scaling Characteristics

Characteristic	Impact	Northwind Interpretation
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Read-Heavy API	JSON serialization + connection setup dominate latency	Most requests are simple reads; CPU rarely becomes the bottleneck.
I/O-Bound Workload	Network saturation occurs before runtime saturation	Python and Go both hit network limits before language/runtime limits.
Single PostgreSQL Instance	Database becomes throughput bottleneck	Two replicas share one DB; DB latency caps total throughput.
Replica Scaling	Improves availability, not throughput	More replicas increase resilience but cannot exceed DB capacity.

C. Practical Implications

Observation	Python	Go	Implication
GIL Impact	Minimal for I/O-bound workloads	No GIL; goroutines scale efficiently	Python remains viable; Go excels under CPU-bound load.
Concurrency Model	Thread-based, lock contention	Atomic operations, lock-free	Go achieves higher concurrency under stress.
Bottleneck Location	Database, not runtime	Database, not runtime	Scaling requires DB improvements, not more replicas.

Future Scaling	Needs read replicas or pooling	Same constraints apply	Horizontal pod scaling alone cannot increase throughput.
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Detailed Explanation

The throughput behavior of the Northwind Logistics delivery service can be understood through two foundational performance models. For a single worker, throughput is approximately the inverse of request latency, meaning that if a request takes 40–50 milliseconds, a single worker can process around 20–25 requests per second. When concurrency is introduced, aggregate throughput becomes limited by the slowest subsystem in the architecture. The theoretical upper bound is expressed as $\min(C/T_{request}, 1/T_{db}, B_{network})$, where the number of concurrent workers, database latency, and network bandwidth collectively determine the maximum achievable throughput. This model is essential for understanding why both the Python and Go implementations behave similarly under realistic load despite their very different concurrency models.

Applied to the Northwind Logistics API, several important characteristics emerge. The service is predominantly read-heavy, with most requests involving simple delivery lookups, JSON serialization, and lightweight response generation. This means that CPU-bound operations are rare, and the runtime itself is not the primary bottleneck. Instead, the workload is I/O-bound, dominated by network latency and database round-trip time. Both Python and Go saturate network bandwidth before hitting runtime limitations, which explains why their throughput converges under typical workloads even though Go's goroutine model is significantly more efficient than Python's thread-based model. Because both replicas share a single PostgreSQL instance, the database becomes the

dominant bottleneck. Even with two Kubernetes replicas, throughput cannot exceed the rate at which PostgreSQL can service queries. This is a classic example of vertical bottlenecking: the database caps throughput regardless of how many application replicas are added.

These observations lead to several practical implications. Python's Global Interpreter Lock (GIL), often cited as a scalability limitation, is far less problematic in this context because the workload is I/O-bound rather than CPU-bound. Python's thread-based concurrency model performs adequately, and lock contention only appears on hot paths such as metrics updates. Go's goroutine-based concurrency model, which uses lock-free atomic operations, would show significant advantages under CPU-bound workloads, but in this project both languages saturate the database before runtime differences matter. This bottleneck analysis justifies the current two-replica architecture: additional replicas improve availability and resilience but do not increase throughput beyond the database's capacity. To scale further, the system would require database-level enhancements such as read replicas, connection pooling, or caching layers.

Finally, Exhibit 8 ties directly into the broader DevOps narrative established across the previous exhibits. Together, Exhibits 1 through 8 form a comprehensive evidence trail demonstrating successful implementation of all six DevOps disciplines: version control, automated testing, CI/CD, infrastructure-as-code, configuration management, containerization, and orchestration. Each exhibit connects directly to specific learning outcomes and implementation details, making the technical report both thorough and verifiable. Exhibit 8 completes this chain by showing that the architecture is not only functional but also grounded in realistic performance modeling and scaling considerations, reinforcing the maturity and completeness of the Northwind Logistics DevOps implementation.

Conclusions

The Northwind Logistics project demonstrates a complete and mature implementation of modern DevOps practices across the full lifecycle of software delivery. The dual-language architecture—Python and Go—provides a unique comparative foundation, showing how different runtimes behave under identical workloads while still integrating seamlessly through containerization and Kubernetes orchestration. The throughput and scaling analysis confirms that the system is correctly optimized for its workload profile: both implementations saturate network and database resources before encountering runtime limitations, validating the decision to deploy two Kubernetes replicas for availability rather than throughput.

The PostgreSQL instance emerges as the primary bottleneck, demonstrating that application-level scaling alone cannot increase throughput without corresponding improvements in the data layer. The CI/CD pipeline is robust and fail-fast, incorporating linting, dual-language testing, Docker image builds, Kubernetes smoke tests, and deployment verification. Infrastructure provisioning through Terraform and configuration management via Ansible further reinforce the project's reproducibility and operational consistency. Collectively, the eight exhibits form a verifiable evidence trail that demonstrates disciplined adherence to DevOps principles and provides a comprehensive, audit-ready technical narrative.

Recommendations

Future DevOps projects should build on this foundation by expanding scalability, observability, security, and automation. The most immediate improvement lies in the data layer: introducing PostgreSQL read replicas, connection pooling, or caching layers such as Redis would alleviate the current throughput bottleneck and allow horizontal scaling of application replicas to translate into

real performance gains. Adopting GitOps tooling such as ArgoCD or FluxCD would elevate deployment governance by making Kubernetes environments fully declarative and version-controlled, improving auditability and rollback safety.

Enhancing observability through Prometheus metrics, Grafana dashboards, distributed tracing with OpenTelemetry, and structured JSON logging would provide deeper visibility into runtime behavior and bottlenecks, enabling more proactive performance tuning. Strengthening the security posture through container image signing, vulnerability scanning, and Kubernetes admission policies would improve supply-chain security and cluster hardening. Integrating performance and load testing tools such as Locust or k6 into CI/CD would validate throughput models under controlled load and detect regressions early. As the system evolves, decomposing the monolithic API into microservices and introducing a service mesh could enable independent scaling, advanced traffic management, and richer telemetry.

Finally, expanding infrastructure-as-code coverage to include managed Kubernetes clusters and cloud networking would enable fully cloud-native deployments, while improving developer experience through pre-commit hooks, local Kubernetes workflows, and automated environment bootstrapping would streamline iteration cycles. Together, these recommendations provide a clear roadmap for advancing the DevOps maturity of future projects.